



# List of TCP and UDP port numbers

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This is a **list of TCP and UDP port numbers** used by protocols for operation of network applications. The Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP) only need one port for duplex, bidirectional traffic. They usually use port numbers that match the services of the corresponding TCP or UDP implementation, if they exist.

The Internet Assigned Numbers Authority (IANA) is responsible for maintaining the official assignments of port numbers for specific uses.<sup>[1]</sup> However, many unofficial uses of both well-known and registered port numbers occur in practice. Similarly, many of the official assignments refer to protocols that were never or are no longer in common use. This article lists port numbers and their associated protocols that have experienced significant uptake.

## Table legend

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Legend of TCP and UDP protocol table cells for port numbers

| Cell       | Description                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes        | Described protocol <i>is</i> assigned by IANA for this port, and <i>is</i> : standardized, specified, or widely used for such.                                                                                  |
| Unofficial | Described protocol <i>is not</i> assigned by IANA for this port, but <i>is</i> : standardized, specified, or widely used for such.                                                                              |
| Assigned   | Described protocol <i>is</i> assigned by IANA for this port, <sup>[2]</sup> but <i>is not</i> : standardized, specified, or widely used for such.                                                               |
| No         | Described protocol <i>is not</i> : assigned by IANA for this port, standardized, specified, or widely used for such.                                                                                            |
| Reserved   | Port is reserved by IANA, <sup>[2]</sup> generally to prevent collision having its previous use removed. <sup>[3][4]</sup> The port number may be available for assignment upon request to IANA. <sup>[3]</sup> |

## Well-known ports

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The port numbers in the range from 0 to 1023 (0 to  $2^{10} - 1$ ) are the *well-known ports* or *system ports*.<sup>[3]</sup> They are used by system processes that provide widely used types of network services. On Unix-like operating systems, a process must execute with superuser privileges to be able to bind a network socket to an IP address using one of the well-known ports.<sup>[5]</sup>

Well-known ports [\[hide\]](#)

| Port | TCP        | UDP        | SCTP                | DCCP     | Description                                                                                                                                                                                     |
|------|------------|------------|---------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | Reserved   |            |                     |          | In programming APIs (not in communication between hosts), requests a system-allocated (dynamic) port <sup>[6]</sup>                                                                             |
| 1    | Yes        | Assigned   |                     |          | TCP Port Service Multiplexer (TCPMUX). Historic. Both TCP and UDP have been assigned to TCPMUX by IANA, <sup>[2]</sup> but by design only TCP is specified. <sup>[7]</sup>                      |
| 2    | Assigned   |            |                     |          | compressnet (Management Utility) <sup>[3]</sup>                                                                                                                                                 |
| 3    | Assigned   |            |                     |          | compressnet (Compression Process) <sup>[3]</sup>                                                                                                                                                |
| 5    | Assigned   |            |                     |          | Remote Job Entry <sup>[8]</sup> was historically using socket 5 in its old socket form, while MIB PIM has identified it as TCP/5 <sup>[9]</sup> and IANA has assigned both TCP and UDP 5 to it. |
| 7    | Yes        |            |                     |          | Echo Protocol <sup>[10][11]</sup>                                                                                                                                                               |
| 9    | Yes        |            | Yes <sup>[12]</sup> | Assigned | Discard Protocol <sup>[13]</sup>                                                                                                                                                                |
|      | No         | Unofficial |                     |          | Wake-on-LAN <sup>[14]</sup>                                                                                                                                                                     |
| 11   | Yes        |            |                     |          | Active Users ( <u>systat</u> service) <sup>[15][16]</sup>                                                                                                                                       |
| 13   | Yes        |            |                     |          | Daytime Protocol <sup>[17]</sup>                                                                                                                                                                |
| 15   | Unofficial | No         |                     |          | Previously <u>netstat</u> service <sup>[2][15]</sup>                                                                                                                                            |
| 17   | Yes        |            |                     |          | Quote of the Day (QOTD) <sup>[18]</sup>                                                                                                                                                         |
| 18   | Yes        |            |                     |          | Message Send Protocol <sup>[19][20]</sup>                                                                                                                                                       |
| 19   | Yes        |            |                     |          | Character Generator Protocol (CHARGEN) <sup>[21]</sup>                                                                                                                                          |
| 20   | Yes        | Assigned   | Yes <sup>[12]</sup> |          | File Transfer Protocol (FTP) data transfer <sup>[11]</sup>                                                                                                                                      |
| 21   | Yes        | Assigned   | Yes <sup>[12]</sup> |          | File Transfer Protocol (FTP) control (command) <sup>[11][12][22][23]</sup>                                                                                                                      |
| 22   | Yes        | Assigned   | Yes <sup>[12]</sup> |          | Secure Shell (SSH), <sup>[11]</sup> secure logins, file transfers ( <u>scp</u> , <u>sftp</u> ) and port forwarding                                                                              |
| 23   | Yes        | Assigned   |                     |          | Telnet protocol—unencrypted text communications <sup>[11][24]</sup>                                                                                                                             |
| 25   | Yes        | Assigned   |                     |          | Simple Mail Transfer Protocol (SMTP), <sup>[11][25]</sup> used for email routing between mail servers                                                                                           |
| 27   | Assigned   |            |                     |          | nsw-fe (NSW User System FE) <sup>[3]</sup>                                                                                                                                                      |
| 28   | Unofficial |            |                     |          | Palo Alto Networks' Panorama High Availability (HA) sync encrypted port. <sup>[26]</sup>                                                                                                        |
| 29   | Assigned   |            |                     |          | msg-icp (MSG ICP) <sup>[3]</sup>                                                                                                                                                                |
| 31   | Assigned   |            |                     |          | msg-auth (MSG Authentication) <sup>[3]</sup>                                                                                                                                                    |
| 33   | Assigned   |            |                     |          | dsp (Display Support Protocol) <sup>[3]</sup>                                                                                                                                                   |

| Port | TCP      | UDP      | SCTP | DCCP | Description                                                                                                                                                                                                                       |
|------|----------|----------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37   | Yes      |          |      |      | <a href="#">Time Protocol</a> <sup>[27]</sup>                                                                                                                                                                                     |
| 38   | Assigned |          |      |      | <a href="#">rap</a> (Route Access Protocol) <sup>[3]</sup>                                                                                                                                                                        |
| 38   | Assigned |          |      |      | <a href="#">rlp</a> (Resource Location Protocol) <sup>[3]</sup>                                                                                                                                                                   |
| 41   | Assigned |          |      |      | <a href="#">graphics</a> (Graphics) <sup>[3]</sup>                                                                                                                                                                                |
| 42   | Assigned | Yes      |      |      | <a href="#">Host Name Server Protocol</a> <sup>[28]</sup>                                                                                                                                                                         |
| 43   | Yes      | Assigned |      |      | <a href="#">WHOIS protocol</a> <sup>[29][30][31]</sup>                                                                                                                                                                            |
| 44   | Assigned |          |      |      | <a href="#">mpm-flags</a> (MPM FLAGS Protocol) <sup>[3]</sup>                                                                                                                                                                     |
| 45   | Assigned |          |      |      | <a href="#">mpm</a> (Message Processing Module [recv]) <sup>[3]</sup>                                                                                                                                                             |
| 46   | Assigned |          |      |      | <a href="#">mpm-snd</a> (MPM [default send]) <sup>[3]</sup>                                                                                                                                                                       |
| 47   | Reserved | Reserved |      |      |                                                                                                                                                                                                                                   |
| 48   | Assigned |          |      |      | <a href="#">auditd</a> (Digital Audit Daemon) <sup>[3]</sup>                                                                                                                                                                      |
| 49   | Yes      |          |      |      | <a href="#">TACACS Login Host protocol</a> . <sup>[32]</sup> <a href="#">TACACS+</a> , still in draft which is an improved but distinct version of TACACS, only uses TCP 49. <sup>[33]</sup>                                      |
| 50   | Assigned |          |      |      | <a href="#">re-mail-ck</a> (Remote Mail Checking Protocol) <sup>[3]</sup>                                                                                                                                                         |
| 51   | Reserved | Reserved |      |      | Historically used for Interface Message Processor logical address management, <sup>[34]</sup> entry has been removed by IANA on 2013-05-25                                                                                        |
| 52   | Assigned |          |      |      | <a href="#">Xerox Network Systems</a> (XNS) Time Protocol. Despite this port being assigned by IANA, the service is meant to work on SPP (ancestor of <a href="#">IPX/SPX</a> ), instead of TCP/IP. <sup>[35]</sup>               |
| 53   | Yes      | Yes      |      |      | <a href="#">Domain Name System</a> (DNS) <sup>[36][11]</sup>                                                                                                                                                                      |
| 54   | Assigned |          |      |      | <a href="#">Xerox Network Systems</a> (XNS) Clearinghouse (Name Server). Despite this port being assigned by IANA, the service is meant to work on SPP (ancestor of <a href="#">IPX/SPX</a> ), instead of TCP/IP. <sup>[35]</sup> |
| 55   | Assigned |          |      |      | <a href="#">isi-gl</a> (ISI Graphics Language) <sup>[3]</sup>                                                                                                                                                                     |
| 56   | Assigned |          |      |      | <a href="#">Xerox Network Systems</a> (XNS) Authentication Protocol. Despite this port being assigned by IANA, the service is meant to work on SPP (ancestor of <a href="#">IPX/SPX</a> ), instead of TCP/IP. <sup>[35]</sup>     |
| 58   | Assigned |          |      |      | <a href="#">Xerox Network Systems</a> (XNS) Mail. Despite this port being assigned by IANA, the service is meant to work on SPP (ancestor of <a href="#">IPX/SPX</a> ), instead of TCP/IP. <sup>[35]</sup>                        |

| Port  | TCP        | UDP        | SCTP                | DCCP | Description                                                                                                                                                                                                                       |
|-------|------------|------------|---------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 61    | Reserved   | Reserved   |                     |      | Historically assigned to the <u>NIFTP-Based Mail protocol</u> , <sup>[37]</sup> but was never documented in the related IEN. <sup>[38]</sup> The port number entry was removed from IANA's registry on 2017-05-18. <sup>[2]</sup> |
| 62    | Assigned   |            |                     |      | acas (ACA Services) <sup>[3]</sup>                                                                                                                                                                                                |
| 63    | Assigned   |            |                     |      | whoispp (whois++) <sup>[3]</sup>                                                                                                                                                                                                  |
| 64    | Assigned   |            |                     |      | covia (Communications Integrator (CI)) <sup>[3]</sup>                                                                                                                                                                             |
| 65    | Assigned   |            |                     |      | tacacs-ds (TACACS-Database Service) <sup>[3]</sup>                                                                                                                                                                                |
| 66    | Assigned   |            |                     |      | sql-net (Oracle SQL*NET) <sup>[3]</sup>                                                                                                                                                                                           |
| 67    | Assigned   | Yes        |                     |      | <u>Bootstrap Protocol (BOOTP) server</u> ; <sup>[11]</sup> also used by <u>Dynamic Host Configuration Protocol (DHCP)</u>                                                                                                         |
| 68    | Assigned   | Yes        |                     |      | <u>Bootstrap Protocol (BOOTP) client</u> ; <sup>[11]</sup> also used by <u>Dynamic Host Configuration Protocol (DHCP)</u>                                                                                                         |
| 69    | Assigned   | Yes        |                     |      | <u>Trivial File Transfer Protocol (TFTP)</u> <sup>[11][39][40][41]</sup>                                                                                                                                                          |
| 70    | Yes        | Assigned   |                     |      | <u>Gopher protocol</u> <sup>[42]</sup>                                                                                                                                                                                            |
| 71–74 | Yes        | Yes        |                     |      | <u>NETRJS protocol</u> <sup>[43][44][45]</sup>                                                                                                                                                                                    |
| 76    | Assigned   |            |                     |      | deos (Distributed External Object Store) <sup>[3]</sup>                                                                                                                                                                           |
| 78    | Assigned   |            |                     |      | vettcp (vettcp) <sup>[3]</sup>                                                                                                                                                                                                    |
| 79    | Yes        | Assigned   |                     |      | <u>Finger protocol</u> <sup>[11][46][47]</sup>                                                                                                                                                                                    |
| 80    | Yes        | Yes        | Yes <sup>[12]</sup> |      | <u>Hypertext Transfer Protocol (HTTP)</u> <sup>[48][49]</sup> uses TCP in versions 1.x and 2. HTTP/3 uses <u>QUIC</u> , <sup>[50]</sup> a transport protocol on top of UDP.                                                       |
| 81    | Unofficial |            |                     |      | <u>TorPark onion routing</u>                                                                                                                                                                                                      |
| 82    | Assigned   |            |                     |      | xfer (XFER Utility) <sup>[3]</sup>                                                                                                                                                                                                |
| 82    |            | Unofficial |                     |      | TorPark control                                                                                                                                                                                                                   |
| 83    | Assigned   |            |                     |      | mit-ml-dev (MIT ML Device) <sup>[3]</sup>                                                                                                                                                                                         |
| 84    | Assigned   |            |                     |      | ctf (Common Trace Facility) <sup>[3]</sup>                                                                                                                                                                                        |
| 85    | Assigned   |            |                     |      | mit-ml-dev (MIT ML Device) <sup>[3]</sup>                                                                                                                                                                                         |
| 86    | Assigned   |            |                     |      | mfcobol (Micro Focus Cobol) <sup>[3]</sup>                                                                                                                                                                                        |
| 88    | Yes        | Yes        |                     |      | <u>Kerberos</u> <sup>[11][51][52]</sup> authentication system                                                                                                                                                                     |
| 89    | Assigned   |            |                     |      | su-mit-tg (SU/MIT Telnet Gateway) <sup>[3]</sup>                                                                                                                                                                                  |
| 90    | Assigned   |            |                     |      | dnsix (DNSIX Security Attribute Token Map) <sup>[3]</sup>                                                                                                                                                                         |
| 90    | Unofficial | Unofficial |                     |      | <u>PointCast (dotcom)</u> <sup>[2]</sup>                                                                                                                                                                                          |

| Port | TCP        | UDP      | SCTP | DCCP | Description                                                                                                                                                                    |
|------|------------|----------|------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 91   | Assigned   |          |      |      | mit-dov (MIT Dover Spooler) <sup>[3]</sup>                                                                                                                                     |
| 92   | Assigned   |          |      |      | npp (Network Printing Protocol) <sup>[3]</sup>                                                                                                                                 |
| 93   | Assigned   |          |      |      | dcp (Device Control Protocol) <sup>[3]</sup>                                                                                                                                   |
| 94   | Assigned   |          |      |      | objcall (Tivoli Object Dispatcher) <sup>[3]</sup>                                                                                                                              |
| 95   | Yes        | Assigned |      |      | SUPDUP, terminal-independent remote login <sup>[53]</sup>                                                                                                                      |
| 96   | Assigned   |          |      |      | dixie (DIXIE Protocol Specification) <sup>[3]</sup>                                                                                                                            |
| 97   | Assigned   |          |      |      | swift-rvf (Swift Remote Virtual File Protocol) <sup>[3]</sup>                                                                                                                  |
| 98   | Assigned   |          |      |      | tacnews (TAC News) <sup>[3]</sup>                                                                                                                                              |
| 99   | Assigned   |          |      |      | metagram (Metagram Relay) <sup>[3]</sup>                                                                                                                                       |
| 101  | Yes        | Assigned |      |      | <u>NIC host name</u> <sup>[54]</sup>                                                                                                                                           |
| 102  | Yes        | Assigned |      |      | <u>ISO Transport Service Access Point (TSAP) Class 0 protocol</u> ; <sup>[55][56]</sup>                                                                                        |
| 104  | Yes        | Yes      |      |      | <u>Digital Imaging and Communications in Medicine (DICOM; also port 11112)</u>                                                                                                 |
| 105  | Yes        | Yes      |      |      | <u>CCSO Nameserver</u> <sup>[57]</sup>                                                                                                                                         |
| 106  | Unofficial | No       |      |      | <u>macOS Server</u> , (macOS) password server <sup>[11]</sup>                                                                                                                  |
| 107  | Yes        | Yes      |      |      | <u>Remote User Telnet Service (RTelnet)</u> <sup>[58]</sup>                                                                                                                    |
| 108  | Yes        | Yes      |      |      | <u>IBM Systems Network Architecture (SNA) gateway access server</u>                                                                                                            |
| 109  | Yes        | Assigned |      |      | <u>Post Office Protocol, version 2 (POP2)</u> <sup>[59]</sup>                                                                                                                  |
| 110  | Yes        | Assigned |      |      | <u>Post Office Protocol, version 3 (POP3)</u> <sup>[11][60][61]</sup>                                                                                                          |
| 111  | Yes        | Yes      |      |      | <u>Open Network Computing Remote Procedure Call (ONC RPC, sometimes referred to as Sun RPC)</u>                                                                                |
| 113  | Yes        | No       |      |      | <u>Ident, authentication service/identification protocol</u> , <sup>[11][62]</sup> used by <u>IRC</u> servers to identify users                                                |
|      | Yes        | Assigned |      |      | <u>Authentication Service (auth)</u> , the predecessor to <i>identification protocol</i> . Used to determine a user's identity of a particular TCP connection. <sup>[63]</sup> |
| 115  | Yes        | Assigned |      |      | <u>Simple File Transfer Protocol</u> <sup>[64]</sup>                                                                                                                           |
| 117  | Yes        | Yes      |      |      | <u>UUCP Mapping Project</u> (path service)                                                                                                                                     |
| 118  | Yes        | Yes      |      |      | <u>Structured Query Language (SQL) Services</u>                                                                                                                                |

| Port | TCP      | UDP      | SCTP | DCCP | Description                                                                                                                                                                               |
|------|----------|----------|------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 119  | Yes      | Assigned |      |      | <u>Network News Transfer Protocol (NNTP)</u> , <sup>[11]</sup> retrieval of newsgroup messages <sup>[65][66]</sup>                                                                        |
| 123  | Assigned | Yes      |      |      | <u>Network Time Protocol (NTP)</u> , used for time synchronization <sup>[11]</sup>                                                                                                        |
| 126  | Yes      | Yes      |      |      | Formerly Unisys Unitary Login, renamed by Unisys to NXEdit. Used by Unisys Programmer's Workbench for Clearpath MCP, an IDE for <u>Unisys MCP software development</u>                    |
| 135  | Yes      | Yes      |      |      | <u>DCE endpoint resolution</u>                                                                                                                                                            |
|      | Yes      | Yes      |      |      | Microsoft EPMAP (End Point Mapper), also known as DCE/RPC Locator service, <sup>[67]</sup> used to remotely manage services including DHCP server, DNS server and WINS. Also used by DCOM |
| 137  | Yes      | Yes      |      |      | <u>NetBIOS Name Service</u> , used for name registration and <u>resolution</u> <sup>[68][69]</sup>                                                                                        |
| 138  | Assigned | Yes      |      |      | <u>NetBIOS Datagram Service</u> <sup>[11][68][69]</sup>                                                                                                                                   |
| 139  | Yes      | Assigned |      |      | <u>NetBIOS Session Service</u> <sup>[68][69]</sup>                                                                                                                                        |
| 143  | Yes      | Assigned |      |      | <u>Internet Message Access Protocol (IMAP)</u> , <sup>[11]</sup> management of <u>electronic mail messages</u> on a server <sup>[70]</sup>                                                |
| 151  | Assigned |          |      |      | <u>HEMS</u>                                                                                                                                                                               |
| 152  | Yes      | Yes      |      |      | <u>Background File Transfer Program (BFTP)</u> <sup>[71]</sup>                                                                                                                            |
| 153  | Yes      | Yes      |      |      | <u>Simple Gateway Monitoring Protocol (SGMP)</u> , a protocol for remote inspection and alteration of gateway management information <sup>[72]</sup>                                      |
| 156  | Yes      | Yes      |      |      | <u>Structured Query Language (SQL) Service</u>                                                                                                                                            |
| 158  | Yes      | Yes      |      |      | <u>Distributed Mail System Protocol (DMSP)</u> , sometimes referred to as Pcmail <sup>[73]</sup>                                                                                          |
| 161  | Assigned | Yes      |      |      | <u>Simple Network Management Protocol (SNMP)</u> <sup>[74][11]</sup>                                                                                                                      |
| 162  | Yes      | Yes      |      |      | <u>Simple Network Management Protocol Trap (SNMPTRAP)</u> <sup>[74][75]</sup>                                                                                                             |
| 165  | Assigned |          |      |      | <u>Xerox</u>                                                                                                                                                                              |
| 169  | Assigned |          |      |      | <u>SEND</u>                                                                                                                                                                               |
| 170  | Yes      | Yes      |      |      | Network <u>PostScript print server</u>                                                                                                                                                    |
| 177  | Yes      | Yes      |      |      | <u>X Display Manager Control Protocol (XDMCP)</u> , used for remote logins to an <u>X Display Manager</u> server <sup>[76]</sup>                                                          |

| Port    | TCP        | UDP      | SCTP                | DCCP | Description                                                                                                                                               |
|---------|------------|----------|---------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 179     | Yes        | Assigned | Yes <sup>[12]</sup> |      | Border Gateway Protocol (BGP), <sup>[77]</sup> used to exchange routing and reachability information among autonomous systems (AS) on the <u>Internet</u> |
| 180     | Assigned   |          |                     |      | <u>ris</u>                                                                                                                                                |
| 194     | Yes        | Yes      |                     |      | <u>Internet Relay Chat (IRC)</u> <sup>[78]</sup>                                                                                                          |
| 199     | Yes        | Yes      |                     |      | <u>SNMP Unix Multiplexer (SMUX)</u> <sup>[79]</sup>                                                                                                       |
| 201     | Yes        | Yes      |                     |      | <u>AppleTalk Routing Maintenance</u>                                                                                                                      |
| 209     | Yes        | Assigned |                     |      | <u>Quick Mail Transfer Protocol</u> <sup>[80]</sup>                                                                                                       |
| 210     | Yes        | Yes      |                     |      | <u>ANSI Z39.50</u>                                                                                                                                        |
| 213     | Yes        | Yes      |                     |      | <u>Internetwork Packet Exchange (IPX)</u>                                                                                                                 |
| 218     | Yes        | Yes      |                     |      | Message posting protocol (MPP)                                                                                                                            |
| 220     | Yes        | Yes      |                     |      | <u>Internet Message Access Protocol (IMAP)</u> , version 3                                                                                                |
| 225–241 | Reserved   | Reserved |                     |      |                                                                                                                                                           |
| 249–255 | Reserved   | Reserved |                     |      |                                                                                                                                                           |
| 259     | Yes        | Yes      |                     |      | Efficient Short Remote Operations (ESRO)                                                                                                                  |
| 262     | Yes        | Yes      |                     |      | Arcisdms                                                                                                                                                  |
| 264     | Yes        | Yes      |                     |      | <u>Border Gateway Multicast Protocol (BGMP)</u>                                                                                                           |
| 280     | Yes        | Yes      |                     |      | http-mgmt                                                                                                                                                 |
| 300     | Unofficial |          |                     |      | <u>ThinLinc</u> Web Access                                                                                                                                |
| 308     | Yes        |          |                     |      | Novastor Online Backup                                                                                                                                    |
| 311     | Yes        | Assigned |                     |      | <u>macOS Server Admin</u> <sup>[11]</sup> (officially <u>AppleShare IP Web administration</u> <sup>[2]</sup> )                                            |
| 312     | Unofficial | No       |                     |      | <u>macOS Xsan administration</u> <sup>[11]</sup>                                                                                                          |
| 318     | Yes        | Yes      |                     |      | PKIX <u>Time Stamp Protocol (TSP)</u>                                                                                                                     |
| 319     |            | Yes      |                     |      | <u>Precision Time Protocol (PTP)</u> event messages                                                                                                       |
| 320     |            | Yes      |                     |      | <u>Precision Time Protocol (PTP)</u> general messages                                                                                                     |
| 350     | Yes        | Yes      |                     |      | <u>Mapping of Airline Traffic over Internet Protocol (MATIP) type A</u>                                                                                   |
| 351     | Yes        | Yes      |                     |      | MATIP type B                                                                                                                                              |
| 356     | Yes        | Yes      |                     |      | cloanto-net-1 (used by Cloanto Amiga Explorer and VMs)                                                                                                    |
| 366     | Yes        | Yes      |                     |      | On-Demand Mail Relay (ODMR)                                                                                                                               |
| 369     | Yes        | Yes      |                     |      | Rpc2portmap                                                                                                                                               |

| Port                    | TCP        | UDP        | SCTP                | DCCP | Description                                                                                                                                                          |
|-------------------------|------------|------------|---------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 370                     | Yes        | Yes        |                     |      | codaaauth2, Coda authentication server                                                                                                                               |
|                         |            | Yes        |                     |      | securecast1, outgoing packets to NAI's SecureCast servers <sup>[81]</sup> As of 2000                                                                                 |
| 371                     | Yes        | Yes        |                     |      | ClearCase albd                                                                                                                                                       |
| 376                     | Yes        | Yes        |                     |      | Amiga Envoy Network Inquiry Protocol                                                                                                                                 |
| 383                     | Yes        | Yes        |                     |      | HP data alarm manager                                                                                                                                                |
| 384                     | Yes        | Yes        |                     |      | A Remote Network Server System                                                                                                                                       |
| 387                     | Yes        | Yes        |                     |      | AURP (AppleTalk Update-based Routing Protocol) <sup>[82]</sup>                                                                                                       |
| 388                     | Yes        | Assigned   |                     |      | Unidata LDM near real-time data distribution protocol <sup>[83][84]</sup>                                                                                            |
| 389                     | Yes        | Assigned   |                     |      | Lightweight Directory Access Protocol (LDAP) <sup>[11]</sup>                                                                                                         |
| 399                     | Yes        | Yes        |                     |      | Digital Equipment Corporation DECnet+ (Phase V) over TCP/IP (RFC1859)                                                                                                |
| 401                     | Yes        | Yes        |                     |      | Uninterruptible power supply (UPS)                                                                                                                                   |
| 427                     | Yes        | Yes        |                     |      | Service Location Protocol (SLP) <sup>[11]</sup>                                                                                                                      |
| 433                     | Yes        | Yes        |                     |      | NNTP, part of Network News Transfer Protocol                                                                                                                         |
| 434                     | Yes        | Yes        |                     |      | Mobile IP Agent (RFC 5944)                                                                                                                                           |
| 443                     | Yes        | Yes        | Yes <sup>[12]</sup> |      | Hypertext Transfer Protocol Secure (HTTPS) <sup>[48][49]</sup> uses TCP in versions 1.x and 2. HTTP/3 uses QUIC, <sup>[50]</sup> a transport protocol on top of UDP. |
| 444                     | Yes        | Yes        |                     |      | Simple Network Paging Protocol (SNPP), RFC 1568                                                                                                                      |
| 445                     | Yes        | Yes        |                     |      | Microsoft-DS (Directory Services) Active Directory, <sup>[85]</sup> Windows shares                                                                                   |
|                         | Yes        | Assigned   |                     |      | Microsoft-DS (Directory Services) SMB <sup>[11]</sup> file sharing                                                                                                   |
| 464                     | Yes        | Yes        |                     |      | Kerberos Change/Set password                                                                                                                                         |
| 465 <sup>[note 1]</sup> | Yes        | No         |                     |      | SMTP over implicit SSL ( <i>obsolete</i> ) <sup>[86]</sup>                                                                                                           |
|                         | Yes        | No         |                     |      | URL Rendezvous Directory for Cisco SSM ( <i>primary usage assignment</i> ) <sup>[87]</sup>                                                                           |
|                         | Yes        | No         |                     |      | Authenticated SMTP <sup>[11]</sup> over TLS/SSL (SMTPS) ( <i>alternative usage assignment</i> ) <sup>[88]</sup>                                                      |
| 475                     | Yes        | Yes        |                     |      | tcpnethaspsrv, Aladdin Knowledge Systems Hasp services                                                                                                               |
| 476–490                 | Unofficial | Unofficial |                     |      | Centro Software ERP ( <a href="https://www.centrossoftware.com/">https://www.centrossoftware.com/</a> ) ports                                                        |

| Port | TCP        | UDP      | SCTP | DCCP | Description                                                                                                                                    |
|------|------------|----------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 491  | Unofficial |          |      |      | <u>GO-Global remote access and application publishing software</u>                                                                             |
| 497  | Yes        | Yes      |      |      | <u>Retrospect</u>                                                                                                                              |
| 500  | Assigned   | Yes      |      |      | <u>Internet Security Association and Key Management Protocol (ISAKMP)</u> / <u>Internet Key Exchange (IKE)</u> <sup>[11]</sup>                 |
| 502  | Yes        | Yes      |      |      | <u>Modbus Protocol</u>                                                                                                                         |
| 504  | Yes        | Yes      |      |      | <u>Citadel</u> , multiservice protocol for dedicated clients for the Citadel groupware system                                                  |
| 510  | Yes        | Yes      |      |      | <u>FirstClass Protocol (FCP)</u> , used by <u>FirstClass</u> client/server groupware system                                                    |
| 512  | Yes        |          |      |      | <u>Rexec</u> , Remote Process Execution                                                                                                        |
|      |            | Yes      |      |      | comsat, together with <u>biff</u>                                                                                                              |
| 513  | Yes        |          |      |      | <u>rlogin</u>                                                                                                                                  |
|      |            | Yes      |      |      | <u>Who</u> <sup>[89]</sup>                                                                                                                     |
| 514  | Unofficial |          |      |      | <u>Remote Shell</u> , used to execute non-interactive commands on a remote system (Remote Shell, rsh, remsh)                                   |
|      | No         | Yes      |      |      | <u>Syslog</u> , <sup>[11]</sup> used for system logging                                                                                        |
| 515  | Yes        | Assigned |      |      | <u>Line Printer Daemon (LPD)</u> , <sup>[11]</sup> print service                                                                               |
| 517  |            | Yes      |      |      | <u>Talk</u>                                                                                                                                    |
| 518  |            | Yes      |      |      | <u>NTalk</u>                                                                                                                                   |
| 520  | Yes        |          |      |      | efs, extended file name server                                                                                                                 |
|      |            | Yes      |      |      | <u>Routing Information Protocol (RIP)</u>                                                                                                      |
| 521  |            | Yes      |      |      | <u>Routing Information Protocol Next Generation (RIPng)</u>                                                                                    |
| 524  | Yes        | Yes      |      |      | <u>NetWare Core Protocol (NCP)</u> is used for a variety things such as access to primary NetWare server resources, Time Synchronization, etc. |
| 525  |            | Yes      |      |      | Timed, <u>Timeserver</u>                                                                                                                       |
| 530  | Yes        | Yes      |      |      | <u>Remote procedure call (RPC)</u>                                                                                                             |
| 532  | Yes        | Assigned |      |      | netnews <sup>[11]</sup>                                                                                                                        |
| 533  |            | Yes      |      |      | netwall, for emergency broadcasts                                                                                                              |
| 540  | Yes        |          |      |      | <u>Unix-to-Unix Copy Protocol (UUCP)</u>                                                                                                       |
| 542  | Yes        | Yes      |      |      | <u>commerce</u> (Commerce Applications)                                                                                                        |
| 543  | Yes        |          |      |      | klogin, <u>Kerberos</u> login                                                                                                                  |
| 544  | Yes        |          |      |      | kshell, Kerberos Remote shell                                                                                                                  |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                      |
|------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 546  | Yes        | Yes        |      |      | DHCPv6 client                                                                                                                                                    |
| 547  | Yes        | Yes        |      |      | DHCPv6 server                                                                                                                                                    |
| 548  | Yes        | Assigned   |      |      | Apple Filing Protocol (AFP) over <a href="#">TCP</a> <sup>[11]</sup>                                                                                             |
| 550  | Yes        | Yes        |      |      | new-rwho, new-who <sup>[89]</sup>                                                                                                                                |
| 554  | Yes        | Yes        |      |      | Real Time Streaming Protocol (RTSP) <sup>[11]</sup>                                                                                                              |
| 556  | Yes        |            |      |      | Remotefs, RFS, rfs_server                                                                                                                                        |
| 560  |            | Yes        |      |      | rmonitor, Remote Monitor                                                                                                                                         |
| 561  |            | Yes        |      |      | monitor                                                                                                                                                          |
| 563  | Yes        | Yes        |      |      | NNTP over <a href="#">TLS/SSL</a> (NNTPS)                                                                                                                        |
| 564  | Unofficial |            |      |      | 9P (Plan 9)                                                                                                                                                      |
| 585  | No         | No         |      |      | Previously assigned for use of Internet Message Access Protocol over <a href="#">TLS/SSL</a> (IMAPS), now deregistered in favour of port 993. <sup>[90]</sup>    |
| 587  | Yes        | Assigned   |      |      | email message submission <sup>[11][91]</sup> (SMTP)                                                                                                              |
| 591  | Yes        |            |      |      | FileMaker 6.0 (and later) Web Sharing (HTTP Alternate, also see port 80)                                                                                         |
| 593  | Yes        | Yes        |      |      | HTTP RPC Ep Map, Remote procedure call over Hypertext Transfer Protocol, often used by Distributed Component Object Model services and Microsoft Exchange Server |
| 601  | Yes        |            |      |      | Reliable Syslog Service — used for system logging                                                                                                                |
| 604  | Yes        |            |      |      | TUNNEL profile, <sup>[92]</sup> a protocol for BEEP peers to form an application layer tunnel                                                                    |
| 623  |            | Yes        |      |      | ASF Remote Management and Control Protocol (ASF-RMCP) & IPMI Remote Management Protocol                                                                          |
| 625  | Unofficial | No         |      |      | Open Directory Proxy (ODProxy) <sup>[11]</sup>                                                                                                                   |
| 631  | Yes        | Yes        |      |      | Internet Printing Protocol (IPP) <sup>[11]</sup>                                                                                                                 |
|      | Unofficial | Unofficial |      |      | Common Unix Printing System (CUPS) administration console (extension to IPP)                                                                                     |
| 635  | Yes        | Yes        |      |      | RLZ DBase                                                                                                                                                        |
| 636  | Yes        | Assigned   |      |      | Lightweight Directory Access Protocol over <a href="#">TLS/SSL</a> (LDAPS) <sup>[11]</sup>                                                                       |
| 639  | Yes        | Yes        |      |      | Multicast Source Discovery Protocol, MSDP                                                                                                                        |
| 641  | Yes        | Yes        |      |      | SupportSoft Nexus Remote Command (control/listening), a proxy gateway connecting remote control traffic                                                          |

| Port | TCP        | UDP      | SCTP | DCCP | Description                                                                                                                                                                                                                                                                          |
|------|------------|----------|------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 643  | Yes        | Yes      |      |      | SANity                                                                                                                                                                                                                                                                               |
| 646  | Yes        | Yes      |      |      | <u>Label Distribution Protocol</u> (LDP), a routing protocol used in <u>MPLS</u> networks                                                                                                                                                                                            |
| 647  | Yes        |          |      |      | <u>DHCP Failover</u> protocol <sup>[93]</sup>                                                                                                                                                                                                                                        |
| 648  | Yes        |          |      |      | Registry Registrar Protocol (RRP) <sup>[94]</sup>                                                                                                                                                                                                                                    |
| 651  | Yes        | Yes      |      |      | IEEE-MMS                                                                                                                                                                                                                                                                             |
| 653  | Yes        | Yes      |      |      | SupportSoft Nexus Remote Command (data), a proxy gateway connecting remote control traffic                                                                                                                                                                                           |
| 654  | Yes        |          |      |      | Media Management System (MMS) Media Management Protocol (MMP) <sup>[95]</sup>                                                                                                                                                                                                        |
| 655  | Yes        | Yes      |      |      | <u>Tinc</u> VPN daemon                                                                                                                                                                                                                                                               |
| 657  | Yes        | Yes      |      |      | <u>IBM RMC</u> (Remote monitoring and Control) protocol, used by <u>System p5 AIX Integrated Virtualization Manager</u> (IVM) <sup>[96]</sup> and <u>Hardware Management Console</u> to connect managed <u>logical partitions</u> (LPAR) to enable dynamic partition reconfiguration |
| 660  | Yes        | Assigned |      |      | macOS Server administration, <sup>[2]</sup> version 10.4 and earlier <sup>[11]</sup>                                                                                                                                                                                                 |
| 666  | Yes        | Yes      |      |      | <u>Doom</u> , the first online <u>first-person shooter</u>                                                                                                                                                                                                                           |
|      | Unofficial |          |      |      | <u>airserv-ng</u> ( <a href="http://www.aircrack-ng.org/doku.php?id=airserv-ng">http://www.aircrack-ng.org/doku.php?id=airserv-ng</a> ), <u>aircrack-ng</u> 's server for remote-controlling wireless devices                                                                        |
| 674  | Yes        |          |      |      | <u>Application Configuration Access Protocol</u> (ACAP)                                                                                                                                                                                                                              |
| 688  | Yes        | Yes      |      |      | REALM-RUSD (ApplianceWare Server Appliance Management Protocol)                                                                                                                                                                                                                      |
| 690  | Yes        | Yes      |      |      | Velneo Application Transfer Protocol (VATP)                                                                                                                                                                                                                                          |
| 691  | Yes        |          |      |      | <u>MS Exchange</u> Routing                                                                                                                                                                                                                                                           |
| 694  | Yes        | Yes      |      |      | <u>Linux-HA</u> high-availability heartbeat                                                                                                                                                                                                                                          |
| 695  | Yes        |          |      |      | <u>IEEE Media Management System over SSL</u> (IEEE-MMS-SSL) <sup>[97]</sup>                                                                                                                                                                                                          |
| 698  |            | Yes      |      |      | <u>Optimized Link State Routing</u> (OLSR)                                                                                                                                                                                                                                           |
| 700  | Yes        |          |      |      | <u>Extensible Provisioning Protocol</u> (EPP), a protocol for communication between <u>domain name registries</u> and <u>registrars</u> (RFC 5734)                                                                                                                                   |
| 701  | Yes        |          |      |      | <u>Link Management Protocol</u> (LMP), <sup>[98]</sup> a protocol that runs between a pair of <u>nodes</u> and is used to manage <u>traffic engineering</u> (TE) <u>links</u>                                                                                                        |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                   |
|------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 702  | Yes        |            |      |      | IRIS <sup>[99][100]</sup> (Internet Registry Information Service) over BEEP (Blocks Extensible Exchange Protocol) <sup>[101]</sup> (RFC 3983) |
| 706  | Yes        |            |      |      | Secure Internet Live Conferencing (SILC)                                                                                                      |
| 711  | Yes        |            |      |      | Cisco Tag Distribution Protocol <sup>[102][103][104]</sup> —being replaced by the MPLS Label Distribution Protocol <sup>[105]</sup>           |
| 712  | Yes        |            |      |      | Topology Broadcast based on Reverse-Path Forwarding routing protocol (TBRPF; RFC 3684)                                                        |
| 749  | Yes        | Yes        |      |      | Kerberos administration <sup>[11]</sup>                                                                                                       |
| 750  |            | Yes        |      |      | kerberos-iv, Kerberos version IV                                                                                                              |
| 751  | Unofficial | Unofficial |      |      | kerberos_master, Kerberos authentication                                                                                                      |
| 752  |            | Unofficial |      |      | passwd_server, Kerberos password (kpasswd) server                                                                                             |
| 753  | Yes        | Yes        |      |      | Reverse Routing Header (RRH) <sup>[106]</sup>                                                                                                 |
|      |            | Unofficial |      |      | userreg_server, Kerberos userreg server                                                                                                       |
| 754  | Yes        | Yes        |      |      | tell send                                                                                                                                     |
|      | Unofficial |            |      |      | krb5_prop, Kerberos v5 slave propagation                                                                                                      |
| 760  | Unofficial | Unofficial |      |      | krbupdate [kreg], Kerberos registration                                                                                                       |
| 782  | Unofficial |            |      |      | Conserver serial-console management server                                                                                                    |
| 783  | Unofficial |            |      |      | SpamAssassin spamd daemon                                                                                                                     |
| 800  | Yes        | Yes        |      |      | mdbs-daemon                                                                                                                                   |
| 802  | Yes        | Yes        |      |      | MODBUS/TCP Security <sup>[107]</sup>                                                                                                          |
| 808  | Unofficial |            |      |      | Microsoft Net.TCP Port Sharing Service                                                                                                        |
| 829  | Yes        | Assigned   |      |      | Certificate Management Protocol <sup>[108]</sup>                                                                                              |
| 830  | Yes        | Yes        |      |      | NETCONF over SSH                                                                                                                              |
| 831  | Yes        | Yes        |      |      | NETCONF over BEEP                                                                                                                             |
| 832  | Yes        | Yes        |      |      | NETCONF for SOAP over HTTPS                                                                                                                   |
| 833  | Yes        | Yes        |      |      | NETCONF for SOAP over BEEP                                                                                                                    |
| 843  | Unofficial |            |      |      | Adobe Flash <sup>[109]</sup>                                                                                                                  |
| 847  | Yes        |            |      |      | DHCP Failover protocol                                                                                                                        |
| 848  | Yes        | Yes        |      |      | Group Domain Of Interpretation (GDOI) protocol                                                                                                |
| 853  | Yes        |            |      |      | DNS over TLS (RFC 7858)                                                                                                                       |
|      |            | Yes        |      |      | DNS over QUIC or DNS over DTLS <sup>[110]</sup>                                                                                               |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                          |
|-----------|------------|------------|------|------|--------------------------------------------------------------------------------------------------------------------------------------|
| 860       | Yes        |            |      |      | <a href="#">iSCSI</a> (RFC 3720)                                                                                                     |
| 861       | Yes        | Yes        |      |      | OWAMP control (RFC 4656)                                                                                                             |
| 862       | Yes        | Yes        |      |      | TWAMP control (RFC 5357)                                                                                                             |
| 873       | Yes        |            |      |      | <a href="#">rsync</a> file synchronization protocol                                                                                  |
| 888       | Unofficial |            |      |      | <a href="#">cddbp</a> , <a href="#">CD DataBase (CDDB)</a> protocol (CDDBP)                                                          |
|           | Unofficial |            |      |      | IBM Endpoint Manager Remote Control                                                                                                  |
| 897       | Unofficial | Unofficial |      |      | <a href="#">Brocade</a> SMI-S RPC                                                                                                    |
| 898       | Unofficial | Unofficial |      |      | Brocade SMI-S RPC SSL                                                                                                                |
| 902       | Unofficial | Unofficial |      |      | <a href="#">VMware ESXi</a> <sup>[111][112]</sup>                                                                                    |
| 903       | Unofficial |            |      |      | <a href="#">VMware ESXi</a> <sup>[111][112]</sup>                                                                                    |
| 953       | Yes        | Reserved   |      |      | <a href="#">BIND</a> remote name daemon control (RNDC) <sup>[113][114]</sup>                                                         |
| 981       | Unofficial |            |      |      | Remote HTTPS management for firewall devices running embedded <a href="#">Check Point VPN-1</a> software <sup>[115]</sup>            |
| 987       |            | Unofficial |      |      | Sony PlayStation Wake On Lan                                                                                                         |
|           | Unofficial |            |      |      | Microsoft Remote Web Workplace, a feature of <a href="#">Windows Small Business Server</a> <sup>[116]</sup>                          |
| 988       | Unofficial |            |      |      | <a href="#">Lustre</a> (file system) <sup>[117]</sup> Protocol (data).                                                               |
| 989       | Yes        | Yes        |      |      | <a href="#">FTPS</a> Protocol (data), <a href="#">FTP</a> over <a href="#">TLS/SSL</a>                                               |
| 990       | Yes        | Yes        |      |      | <a href="#">FTPS</a> Protocol (control), <a href="#">FTP</a> over <a href="#">TLS/SSL</a>                                            |
| 991       | Yes        | Yes        |      |      | <a href="#">Netnews</a> Administration System (NAS) <sup>[118]</sup>                                                                 |
| 992       | Yes        | Yes        |      |      | <a href="#">Telnet</a> protocol over <a href="#">TLS/SSL</a>                                                                         |
| 993       | Yes        | Assigned   |      |      | <a href="#">Internet Message Access Protocol</a> over <a href="#">TLS/SSL</a> (IMAPS) <sup>[11]</sup>                                |
| 994       | Reserved   | Reserved   |      |      | Previously assigned to <a href="#">Internet Relay Chat</a> over <a href="#">TLS/SSL</a> (IRCS), but was not used in common practice. |
| 995       | Yes        | Yes        |      |      | <a href="#">Post Office Protocol 3</a> over <a href="#">TLS/SSL</a> (POP3S) <sup>[11]</sup>                                          |
| 1010      | Unofficial |            |      |      | <a href="#">ThinLinc</a> web-based administration interface <sup>[119]</sup>                                                         |
| 1011–1020 | Reserved   | Reserved   |      |      |                                                                                                                                      |
| 1023      | Reserved   | Reserved   |      |      | <sup>[2]</sup>                                                                                                                       |
|           | Unofficial | Unofficial |      |      | <a href="#">z/OS</a> Network File System (NFS) (potentially ports 991–1023) <sup>[78][79][120]</sup>                                 |

## Registered ports

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The range of port numbers from 1024 to 49151 ( $2^{10}$  to  $2^{14} + 2^{15} - 1$ ) are the registered ports. They are assigned by IANA for specific service upon application by a requesting entity.<sup>[2]</sup> On most systems, registered ports can be used without superuser privileges.

Registered ports [\[hide\]](#)

| Port | TCP                                           | UDP                  | SCTP | DCCP | Description                                                                                                         |
|------|-----------------------------------------------|----------------------|------|------|---------------------------------------------------------------------------------------------------------------------|
| 1024 | Reserved                                      | Reserved             |      |      | Reserved                                                                                                            |
| 1025 | Yes                                           | Yes                  |      |      | <u>Teradata database management system</u> (Teradata) server                                                        |
| 1027 | Reserved                                      |                      |      |      | Reserved                                                                                                            |
|      |                                               | Yes                  |      |      | Native IPv6 behind IPv4-to-IPv4 NAT Customer Premises Equipment (6a44) <sup>[121]</sup>                             |
| 1028 | Reserved <sup>[2]</sup>                       |                      |      |      |                                                                                                                     |
| 1029 | Unofficial                                    |                      |      |      | Microsoft <u>DCOM</u> services                                                                                      |
| 1058 | Yes                                           | Yes                  |      |      | <u>nim</u> , <u>IBM AIX Network Installation Manager</u> (NIM)                                                      |
| 1059 | Yes                                           | Yes                  |      |      | <u>nimreg</u> , IBM AIX Network Installation Manager (NIM)                                                          |
| 1080 | Yes                                           | Yes                  |      |      | <u>SOCKS</u> proxy                                                                                                  |
| 1085 | Yes                                           | Yes                  |      |      | <u>WebObjects</u> <sup>[11]</sup>                                                                                   |
| 1098 | Yes                                           | Yes                  |      |      | <u>rmiactivation</u> , Java <u>remote method invocation</u> (RMI) activation                                        |
| 1099 | Yes                                           | Assigned             |      |      | <u>rmiregistry</u> , Java <u>remote method invocation</u> (RMI) registry                                            |
| 1109 | Reserved                                      | Reserved             |      |      | Reserved                                                                                                            |
| 1113 | Assigned <sup>[note 2]</sup> <sup>[122]</sup> | Yes <sup>[123]</sup> |      |      | <u>Licklider Transmission Protocol</u> (LTP) delay tolerant networking protocol                                     |
| 1119 | Yes                                           | Yes                  |      |      | <u>Battle.net chat/game protocol</u> , used by <u>Blizzard's games</u> <sup>[124]</sup>                             |
| 1167 | Yes                                           | Yes                  | Yes  |      | Cisco <u>IP SLA</u> (Service Assurance Agent)                                                                       |
| 1194 | Yes                                           | Yes                  |      |      | <u>OpenVPN</u>                                                                                                      |
| 1198 | Yes                                           | Yes                  |      |      | The <u>cajo project</u> Free dynamic transparent distributed computing in Java                                      |
| 1212 | Unofficial                                    | Unofficial           |      |      | Equalsocial <u>Fediverse</u> protocol                                                                               |
| 1214 | Yes                                           | Yes                  |      |      | <u>Kazaa</u>                                                                                                        |
| 1220 | Yes                                           | Assigned             |      |      | <u>QuickTime Streaming Server</u> administration <sup>[11]</sup>                                                    |
| 1234 | Yes                                           | Yes                  |      |      | <u>Infoseek</u> search agent                                                                                        |
|      |                                               | Unofficial           |      |      | <u>VLC media player</u> default port for UDP/RTP stream                                                             |
| 1241 | Unofficial                                    | Unofficial           |      |      | <u>Nessus Security Scanner</u>                                                                                      |
| 1270 | Yes                                           | Yes                  |      |      | Microsoft <u>System Center Operations Manager</u> (SCOM) (formerly Microsoft <u>Operations Manager</u> (MOM)) agent |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                              |
|------|------------|------------|------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1293 | Yes        | Yes        |      |      | Internet Protocol Security (IPSec)                                                                                                                                                                                       |
| 1311 | Yes        | Yes        |      |      | Windows RxMon . exe                                                                                                                                                                                                      |
|      | Unofficial |            |      |      | Dell <u>OpenManage</u> HTTPS <sup>[125]</sup>                                                                                                                                                                            |
| 1314 | Unofficial |            |      |      | <u>Festival Speech Synthesis System</u> server <sup>[126]</sup>                                                                                                                                                          |
| 1319 | Yes        | Yes        |      |      | AMX ICSP (Protocol for communications with AMX control systems devices)                                                                                                                                                  |
| 1337 | Yes        | Yes        |      |      | <u>Men&amp;Mice DNS</u> <sup>[127]</sup>                                                                                                                                                                                 |
|      | Unofficial |            |      |      | <u>Strapi</u> <sup>[128]</sup>                                                                                                                                                                                           |
|      | Unofficial |            |      |      | <u>Sails.js</u> default port <sup>[129]</sup>                                                                                                                                                                            |
| 1341 | Yes        | Yes        |      |      | Qubes ( <u>Manufacturing Execution System</u> )                                                                                                                                                                          |
| 1344 | Yes        | Yes        |      |      | <u>Internet Content Adaptation Protocol</u>                                                                                                                                                                              |
| 1352 | Yes        | Yes        |      |      | IBM <u>Lotus Notes/Domino</u> (RPC) protocol                                                                                                                                                                             |
| 1360 | Yes        | Yes        |      |      | <u>Mimer SQL</u>                                                                                                                                                                                                         |
| 1414 | Yes        | Yes        |      |      | IBM <u>WebSphere MQ</u> (formerly known as MQSeries)                                                                                                                                                                     |
| 1417 | Yes        | Yes        |      |      | <u>Timbuktu</u> Service 1 Port                                                                                                                                                                                           |
| 1418 | Yes        | Yes        |      |      | Timbuktu Service 2 Port                                                                                                                                                                                                  |
| 1419 | Yes        | Yes        |      |      | Timbuktu Service 3 Port                                                                                                                                                                                                  |
| 1420 | Yes        | Yes        |      |      | Timbuktu Service 4 Port                                                                                                                                                                                                  |
| 1431 | Yes        |            |      |      | Reverse Gossip Transport Protocol (RGTP), used to access a General-purpose Reverse-Ordered Gossip Gathering System (GROGGS) bulletin board, such as that implemented on the <u>Cambridge University's Phoenix system</u> |
| 1433 | Yes        | Yes        |      |      | Microsoft SQL Server database management system (MSSQL) server                                                                                                                                                           |
| 1434 | Yes        | Yes        |      |      | Microsoft SQL Server database management system (MSSQL) monitor                                                                                                                                                          |
| 1476 | Yes        | Yes        |      |      | WiFi Pineapple Hak5.                                                                                                                                                                                                     |
| 1481 | Yes        | Yes        |      |      | AIRS data interchange.                                                                                                                                                                                                   |
| 1492 | Unofficial |            |      |      | <i>Sid Meier's CivNet</i> , a multiplayer remake of the original <i>Sid Meier's Civilization</i> game                                                                                                                    |
| 1494 | Unofficial | Unofficial |      |      | Citrix <u>Independent Computing Architecture</u> (ICA) <sup>[130]</sup>                                                                                                                                                  |
| 1500 | Unofficial |            |      |      | IBM <u>Tivoli Storage Manager</u> server <sup>[131]</sup>                                                                                                                                                                |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                  |
|-----------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------|
| 1501      | Unofficial |            |      |      | IBM Tivoli Storage Manager client scheduler <sup>[131]</sup>                                                                 |
| 1503      | Unofficial | Unofficial |      |      | Windows Live Messenger (Whiteboard and Application Sharing) <sup>[132]</sup>                                                 |
| 1512      | Yes        | Yes        |      |      | Microsoft's Windows Internet Name Service (WINS)                                                                             |
| 1513      | Unofficial | Unofficial |      |      | Garena game client                                                                                                           |
| 1521      | Yes        | Yes        |      |      | nCUBE License Manager                                                                                                        |
|           | Unofficial |            |      |      | Oracle database default listener, in future releases <sup>[133]</sup> official port 2483 (TCP/IP) and 2484 (TCP/IP with SSL) |
| 1524      | Yes        | Yes        |      |      | ingreslock, ingres                                                                                                           |
| 1527      | Yes        | Yes        |      |      | Oracle Net Services, formerly known as SQL*Net <sup>[134]</sup>                                                              |
|           | Unofficial |            |      |      | Apache Derby Network Server <sup>[135]</sup>                                                                                 |
| 1533      | Yes        | Yes        |      |      | IBM Sametime Virtual Places Chat                                                                                             |
| 1534      | No         | Unofficial |      |      | Eclipse Target Communication Framework <sup>[136]</sup>                                                                      |
| 1540      | Unofficial | Unofficial |      |      | 1C:Enterprise server agent (ragent) <sup>[137][138]</sup>                                                                    |
| 1541      | Unofficial | Unofficial |      |      | 1C:Enterprise master cluster manager (rmngr) <sup>[137]</sup>                                                                |
| 1542      | Unofficial | Unofficial |      |      | 1C:Enterprise configuration repository server <sup>[137]</sup>                                                               |
| 1545      | Unofficial | Unofficial |      |      | 1C:Enterprise cluster administration server (RAS) <sup>[137]</sup>                                                           |
| 1547      | Yes        | Yes        |      |      | Laplink                                                                                                                      |
| 1550      | Unofficial | Unofficial |      |      | 1C:Enterprise debug server <sup>[137]</sup>                                                                                  |
|           | Unofficial |            |      |      | Gadu-Gadu (direct client-to-client)                                                                                          |
| 1560–1590 | Unofficial | Unofficial |      |      | 1C:Enterprise cluster working processes <sup>[137]</sup>                                                                     |
| 1581      | Yes        | Yes        |      |      | MIL STD 2045-47001 VMF                                                                                                       |
|           | Unofficial |            |      |      | IBM Tivoli Storage Manager web client <sup>[131]</sup>                                                                       |
| 1582–1583 | Unofficial |            |      |      | IBM Tivoli Storage Manager server web interface <sup>[131]</sup>                                                             |
| 1583      | Unofficial |            |      |      | Pervasive PSQL <sup>[139]</sup>                                                                                              |
| 1589      | Yes        | Yes        |      |      | Cisco VLAN Query Protocol (VQP)                                                                                              |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                                                                                                                |
|-----------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1604      | Unofficial | Unofficial |      |      | <u>DarkComet</u> remote administration tool (RAT)                                                                                                                                                                                                                                                          |
| 1626      | Unofficial |            |      |      | <u>iSketch</u> <sup>[140]</sup>                                                                                                                                                                                                                                                                            |
| 1627      | Unofficial |            |      |      | <u>iSketch</u> <sup>[140]</sup>                                                                                                                                                                                                                                                                            |
| 1628      | Yes        | Yes        |      |      | <u>LonTalk</u> normal                                                                                                                                                                                                                                                                                      |
| 1629      | Yes        | Yes        |      |      | <u>LonTalk</u> urgent                                                                                                                                                                                                                                                                                      |
| 1645      | No         | Unofficial |      |      | Early deployment of <u>RADIUS</u> before RFC standardization was done using UDP port number 1645. Enabled for compatibility reasons by default on <u>Cisco</u> and <u>Juniper Networks</u> RADIUS servers. <sup>[141]</sup> Official port is 1812. TCP port 1645 <b>MUST NOT</b> be used. <sup>[142]</sup> |
| 1646      | No         | Unofficial |      |      | Old radacct port, RADIUS accounting protocol. Enabled for compatibility reasons by default on <u>Cisco</u> and <u>Juniper Networks</u> RADIUS servers. <sup>[141]</sup> Official port is 1813. TCP port 1646 <b>MUST NOT</b> be used. <sup>[142]</sup>                                                     |
| 1666      | Unofficial |            |      |      | <u>Perforce</u> <sup>[143]</sup>                                                                                                                                                                                                                                                                           |
| 1677      | Yes        | Yes        |      |      | <u>Novell GroupWise</u> clients in client/server access mode                                                                                                                                                                                                                                               |
| 1688      | Unofficial |            |      |      | Microsoft <u>Key Management Service</u> (KMS) for Windows Activation <sup>[144]</sup>                                                                                                                                                                                                                      |
| 1701      | Yes        | Yes        |      |      | <u>Layer 2 Forwarding Protocol</u> (L2F)                                                                                                                                                                                                                                                                   |
|           | Assigned   | Yes        |      |      | <u>Layer 2 Tunneling Protocol</u> (L2TP) <sup>[11]</sup>                                                                                                                                                                                                                                                   |
| 1707      | Yes        | Yes        |      |      | <u>Windward Studios</u> games (vdmplay)                                                                                                                                                                                                                                                                    |
|           |            | Unofficial |      |      | L2TP/IPsec, for establish an initial connection <sup>[145]</sup>                                                                                                                                                                                                                                           |
| 1714–1764 | Unofficial | Unofficial |      |      | <u>KDE Connect</u> <sup>[146]</sup>                                                                                                                                                                                                                                                                        |
| 1716      |            | Unofficial |      |      | <u>America's Army</u> , a massively multiplayer online game (MMO) <sup>[147]</sup>                                                                                                                                                                                                                         |
| 1719      | Yes        | Yes        |      |      | H.323 registration and alternate communication                                                                                                                                                                                                                                                             |
| 1720      | Yes        | Yes        |      |      | H.323 call signaling                                                                                                                                                                                                                                                                                       |
| 1723      | Yes        | Assigned   |      |      | <u>Point-to-Point Tunneling Protocol</u> (PPTP) <sup>[11]</sup>                                                                                                                                                                                                                                            |
| 1755      | Yes        | Yes        |      |      | <u>Microsoft Media Services</u> (MMS, ms-streaming)                                                                                                                                                                                                                                                        |
| 1761      | Unofficial | Unofficial |      |      | <u>Novell ZENworks</u> <sup>[148][149]</sup>                                                                                                                                                                                                                                                               |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                    |
|------|------------|------------|------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1776 | Yes        |            |      |      | <u>Emergency management information system</u>                                                                                                                                 |
| 1783 | Reserved   |            |      |      | "Decomissioned [ <i>sic</i> ] Port 04/14/00, ms" <sup>[2]</sup>                                                                                                                |
| 1801 | Yes        | Yes        |      |      | <u>Microsoft Message Queuing</u>                                                                                                                                               |
| 1812 | Yes        | Yes        |      |      | <u>RADIUS</u> authentication protocol, radius                                                                                                                                  |
| 1813 | Yes        | Yes        |      |      | <u>RADIUS</u> accounting protocol, radius-acct                                                                                                                                 |
| 1863 | Yes        | Yes        |      |      | <u>Microsoft Notification Protocol (MSNP)</u> , used by the <u>Microsoft Messenger service</u> and a number of instant messaging <u>Messenger clients</u>                      |
| 1880 | Unofficial |            |      |      | <u>Node-RED</u> <sup>[150]</sup>                                                                                                                                               |
| 1883 | Yes        | Yes        |      |      | <u>MQTT</u> (formerly <u>MQ Telemetry Transport</u> )                                                                                                                          |
| 1900 | Assigned   | Yes        |      |      | <u>Simple Service Discovery Protocol (SSDP)</u> , <sup>[11]</sup> discovery of <u>UPnP</u> devices                                                                             |
| 1935 | Yes        | Yes        |      |      | <u>Macromedia Flash Communications Server MX</u> , the precursor to <u>Adobe Flash Media Server</u> before <u>Macromedia's</u> acquisition by <u>Adobe</u> on December 3, 2005 |
|      | Unofficial | Unofficial |      |      | <u>Real Time Messaging Protocol (RTMP)</u> , primarily used in <u>Adobe Flash</u> <sup>[151]</sup>                                                                             |
| 1965 | Unofficial | No         |      |      | <u>Gemini</u> , a lightweight, collaboratively designed protocol, striving to fill the gap between <u>Gopher</u> and <u>HTTP</u> <sup>[152]</sup>                              |
| 1967 |            | Unofficial |      |      | <u>Cisco IOS IP Service Level Agreements (IP SLAs) Control Protocol</u>                                                                                                        |
| 1972 | Yes        | Yes        |      |      | <u>InterSystems Caché</u> , and <u>InterSystems IRIS</u> versions 2020.3 and later                                                                                             |
| 1984 | Yes        | Yes        |      |      | <u>Big Brother</u>                                                                                                                                                             |
| 1985 | Assigned   | Yes        |      |      | <u>Cisco Hot Standby Router Protocol (HSRP)</u> <sup>[153]</sup>                                                                                                               |
| 1998 | Yes        | Yes        |      |      | <u>Cisco X.25 over TCP (XOT)</u> service                                                                                                                                       |
| 2000 | Yes        | Yes        |      |      | <u>Cisco Skinny Client Control Protocol (SCCP)</u>                                                                                                                             |
| 2010 | Unofficial |            |      |      | <u>Artemis: Spaceship Bridge Simulator</u> <sup>[154]</sup>                                                                                                                    |
| 2033 | Unofficial | Unofficial |      |      | <u>Civilization IV</u> multiplayer <sup>[155]</sup>                                                                                                                            |
| 2049 | Yes        | Yes        | Yes  |      | <u>Network File System (NFS)</u> <sup>[11]</sup>                                                                                                                               |
| 2056 | Unofficial | Unofficial |      |      | <u>Civilization IV</u> multiplayer <sup>[155]</sup>                                                                                                                            |
| 2080 | Yes        | Yes        |      |      | <u>Autodesk NLM (FLEXlm)</u>                                                                                                                                                   |

| Port      | TCP        | UDP | SCTP | DCCP | Description                                                                                                             |
|-----------|------------|-----|------|------|-------------------------------------------------------------------------------------------------------------------------|
| 2082      | Unofficial |     |      |      | cPanel default <sup>[156]</sup>                                                                                         |
| 2083      | Yes        | Yes |      |      | Secure <u>RADIUS</u> Service (radsec)                                                                                   |
|           | Unofficial |     |      |      | cPanel default <u>SSL</u> <sup>[156]</sup>                                                                              |
| 2086      | Yes        | Yes |      |      | <u>GNUnet</u>                                                                                                           |
|           | Unofficial |     |      |      | <u>WebHost Manager</u> default <sup>[156]</sup>                                                                         |
| 2087      | Unofficial |     |      |      | <u>WebHost Manager</u> default <u>SSL</u> <sup>[156]</sup>                                                              |
| 2095      | Yes        |     |      |      | cPanel default web mail <sup>[156]</sup>                                                                                |
| 2096      | Unofficial |     |      |      | cPanel default <u>SSL</u> web mail <sup>[156]</sup>                                                                     |
| 2100      | Unofficial |     |      |      | <u>Warzone 2100</u> multiplayer                                                                                         |
| 2101      | Unofficial |     |      |      | <u>Networked Transport of RTCM via Internet Protocol</u> (NTRIP)                                                        |
| 2102      | Yes        | Yes |      |      | <u>Zephyr Notification Service</u> server                                                                               |
| 2103      | Yes        | Yes |      |      | <u>Zephyr Notification Service</u> serv-hm connection                                                                   |
| 2104      | Yes        | Yes |      |      | <u>Zephyr Notification Service</u> hostmanager                                                                          |
| 2123      | Yes        | Yes |      |      | <u>GTP</u> control messages (GTP-C)                                                                                     |
| 2142      | Yes        | Yes |      |      | <u>TDMoIP</u> (TDM over IP)                                                                                             |
| 2152      | Yes        | Yes |      |      | <u>GTP</u> user data messages (GTP-U)                                                                                   |
| 2159      | Yes        | Yes |      |      | <u>GDB remote debug port</u>                                                                                            |
| 2181      | Yes        | Yes |      |      | EForward-document transport system                                                                                      |
|           | Unofficial |     |      |      | <u>Apache ZooKeeper</u> default client port                                                                             |
| 2195      | Unofficial |     |      |      | <u>Apple Push Notification Service</u> , binary, gateway. <sup>[11][157]</sup> Deprecated March 2021. <sup>[158]</sup>  |
| 2196      | Unofficial |     |      |      | <u>Apple Push Notification Service</u> , binary, feedback. <sup>[11][157]</sup> Deprecated March 2021. <sup>[158]</sup> |
| 2197      | Unofficial |     |      |      | <u>Apple Push Notification Service</u> , <u>HTTP/2</u> , <u>JSON</u> -based <u>API</u> .                                |
| 2210      | Yes        | Yes |      |      | <u>NOAAPORT</u> Broadcast Network                                                                                       |
| 2211      | Yes        | Yes |      |      | <u>EMWIN</u>                                                                                                            |
| 2221      | Unofficial |     |      |      | <u>ESET</u> anti-virus updates <sup>[159]</sup>                                                                         |
| 2222      | Yes        | Yes |      |      | <u>EtherNet/IP</u> implicit messaging for IO data                                                                       |
|           | Unofficial |     |      |      | <u>DirectAdmin</u> Access <sup>[160]</sup>                                                                              |
| 2222-2226 | Yes        |     |      |      | ESET Remote administrator <sup>[159]</sup>                                                                              |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                       |
|------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 2240 | Yes        | Yes        |      |      | General Dynamics Remote Encryptor Configuration Information Protocol (RECIPe)                                                     |
| 2261 | Yes        | Yes        |      |      | <u>CoMotion</u> master                                                                                                            |
| 2262 | Yes        | Yes        |      |      | CoMotion backup                                                                                                                   |
| 2302 |            | Unofficial |      |      | <u>ArmA</u> multiplayer <sup>[161]</sup>                                                                                          |
|      |            | Unofficial |      |      | Halo: Combat Evolved multiplayer host <sup>[162]</sup>                                                                            |
| 2303 |            | Unofficial |      |      | ArmA multiplayer ( <i>default port for game +1</i> ) <sup>[161]</sup>                                                             |
|      |            | Unofficial |      |      | Halo: Combat Evolved multiplayer listener <sup>[162]</sup>                                                                        |
| 2305 |            | Unofficial |      |      | ArmA multiplayer ( <i>default port for game +3</i> ) <sup>[161]</sup>                                                             |
| 2351 | Unofficial |            |      |      | <u>AIM</u> game LAN network port                                                                                                  |
| 2368 | Unofficial |            |      |      | <u>Ghost</u> (blogging platform) <sup>[163]</sup>                                                                                 |
| 2369 | Unofficial |            |      |      | Default for BMC Control-M/Server Configuration Agent                                                                              |
| 2370 | Unofficial |            |      |      | Default for BMC Control-M/Server, to allow the Control-M/Enterprise Manager to connect to the Control-M/Server                    |
| 2372 | Unofficial |            |      |      | Default for <u>K9 Web Protection</u> /parental controls, content filtering agent                                                  |
| 2375 | Yes        | Reserved   |      |      | <u>Docker</u> REST API (plain)                                                                                                    |
| 2376 | Yes        | Reserved   |      |      | Docker REST API (SSL)                                                                                                             |
| 2377 | Yes        | Reserved   |      |      | Docker Swarm cluster management communications <sup>[164]</sup>                                                                   |
| 2379 | Yes        | Reserved   |      |      | CoreOS <u>etcd</u> client communication                                                                                           |
|      | Unofficial |            |      |      | <u>KGS Go Server</u> <sup>[165]</sup>                                                                                             |
| 2380 | Yes        | Reserved   |      |      | CoreOS etcd server communication                                                                                                  |
| 2389 | Assigned   |            |      |      | OpenView Session Mgr                                                                                                              |
| 2399 | Yes        | Yes        |      |      | <u>FileMaker</u> Data Access Layer (ODBC/JDBC)                                                                                    |
| 2401 | Yes        | Yes        |      |      | CVS version control system password-based server                                                                                  |
| 2404 | Yes        | Yes        |      |      | IEC 60870-5-104, used to send electric power telecontrol messages between two systems via directly connected <u>data circuits</u> |
| 2424 | Unofficial |            |      |      | <u>OrientDB</u> database listening for binary client connections <sup>[166]</sup>                                                 |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                               |
|-----------|------------|------------|------|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 2427      | Yes        | Yes        |      |      | Media Gateway Control Protocol (MGCP) media gateway                                                                                       |
| 2447      | Yes        | Yes        |      |      | ovwdb— <u>OpenView Network Node Manager</u> (NNM) daemon                                                                                  |
| 2456      | Unofficial | Unofficial |      |      | <u>Valheim</u>                                                                                                                            |
| 2459      | Yes        | Yes        |      |      | <u>XRPL</u>                                                                                                                               |
| 2480      | Unofficial |            |      |      | <u>OrientDB</u> database listening for HTTP client connections <sup>[166]</sup>                                                           |
| 2483      | Yes        | Yes        |      |      | Oracle database listening for insecure client connections to the listener, replaces port 1521                                             |
| 2484      | Yes        | Yes        |      |      | Oracle database listening for <u>SSL</u> client connections to the listener                                                               |
| 2500      | Unofficial | Unofficial |      |      | NetFS communication <sup>[167]</sup>                                                                                                      |
| 2501      |            | Unofficial |      |      | NetFS probe                                                                                                                               |
| 2535      | Yes        | Yes        |      |      | Multicast Address Dynamic Client Allocation Protocol (MADCAP). <sup>[168]</sup> All standard messages are UDP datagrams. <sup>[169]</sup> |
| 2541      | Yes        | Yes        |      |      | <u>LonTalk</u> /IP                                                                                                                        |
| 2546–2548 | Yes        | Yes        |      |      | <u>EVault</u> data protection services                                                                                                    |
| 2593      | Unofficial | Unofficial |      |      | <i>Ultima Online</i> servers                                                                                                              |
| 2598      | Unofficial |            |      |      | Citrix Independent Computing Architecture (ICA) with Session Reliability; port 1494 without session reliability <sup>[130]</sup>          |
| 2599      | Unofficial | Unofficial |      |      | <i>Ultima Online</i> servers                                                                                                              |
| 2628      | Yes        | Yes        |      |      | <u>DICT</u> <sup>[170]</sup>                                                                                                              |
| 2638      | Yes        | Yes        |      |      | <u>SQL Anywhere</u> database server <sup>[171][172]</sup>                                                                                 |
| 2710      | Unofficial | Unofficial |      |      | XBT Tracker. <sup>[173]</sup> UDP tracker extension is considered experimental. <sup>[174]</sup>                                          |
| 2727      | Yes        | Yes        |      |      | Media Gateway Control Protocol (MGCP) media gateway controller (call agent)                                                               |
| 2775      | Yes        | Yes        |      |      | <u>Short Message Peer-to-Peer</u> (SMPP)                                                                                                  |
| 2809      | Yes        | Yes        |      |      | corbaloc:iiop URL, per the <u>CORBA</u> 3.0.3 specification                                                                               |
| 2811      | Yes        | Yes        |      |      | gsi ftp, per the <u>GridFTP</u> specification                                                                                             |
| 2827      | Unofficial |            |      |      | <u>I2P BOB Bridge</u> <sup>[175]</sup>                                                                                                    |
| 2944      | Yes        | Yes        |      |      | <u>Megaco</u> text H.248                                                                                                                  |
| 2945      | Yes        | Yes        |      |      | Megaco binary (ASN.1) H.248                                                                                                               |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                     |
|-----------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------|
| 2947      | Yes        | Yes        |      |      | <u>gpsd</u> , GPS daemon                                                                                        |
| 2948–2949 | Yes        | Yes        |      |      | <u>WAP push</u> <u>Multimedia Messaging Service</u> (MMS)                                                       |
| 2967      | Yes        | Yes        |      |      | <u>Symantec System Center</u> agent (SSC-AGENT)                                                                 |
| 3000      | Unofficial |            |      |      | <u>Ruby on Rails</u> development default <sup>[176]</sup>                                                       |
|           | Unofficial |            |      |      | <u>Meteor</u> development default <sup>[177]</sup>                                                              |
|           | Unofficial | Unofficial |      |      | <u>Resilio Sync</u> , <sup>[178]</sup> spun from BitTorrent Sync.                                               |
|           | Unofficial |            |      |      | Create React App, script to create single-page <u>React</u> applications <sup>[179]</sup>                       |
|           | Unofficial |            |      |      | Gogs (self-hosted GIT service) <sup>[180]</sup>                                                                 |
|           | Unofficial |            |      |      | <u>Grafana</u> <sup>[181]</sup>                                                                                 |
| 3001      | Yes        | No         |      |      | Honeywell Prowatch <sup>[182]</sup>                                                                             |
| 3004      | Unofficial |            |      |      | <u>iSync</u> <sup>[11]</sup>                                                                                    |
| 3010      | Yes        | Yes        |      |      | KWS Connector                                                                                                   |
| 3020      | Yes        | Yes        |      |      | Common Internet File System (CIFS). See also port 445 for <u>Server Message Block</u> (SMB), a dialect of CIFS. |
| 3050      | Yes        | Yes        |      |      | gds-db ( <u>Interbase</u> / <u>Firebird</u> databases)                                                          |
| 3052      | Yes        | Yes        |      |      | <u>APC PowerChute Network</u>                                                                                   |
| 3074      | Yes        | Yes        |      |      | Xbox LIVE and <u>Games for Windows – Live</u>                                                                   |
| 3101      | Unofficial |            |      |      | <u>BlackBerry Enterprise Server</u> communication protocol <sup>[183]</sup>                                     |
| 3128      | Unofficial | No         |      |      | <u>Squid</u> caching web proxy <sup>[184]</sup>                                                                 |
| 3225      | Yes        | Yes        |      |      | <u>Fibre Channel over IP</u> (FCIP)                                                                             |
| 3233      | Yes        | Yes        |      |      | <u>WhiskerControl</u> research control protocol                                                                 |
| 3260      | Yes        | Yes        |      |      | <u>iSCSI</u>                                                                                                    |
| 3268      | Yes        | Yes        |      |      | msft-gc, Microsoft Global Catalog (LDAP service which contains data from <u>Active Directory</u> forests)       |
| 3269      | Yes        | Yes        |      |      | msft-gc-ssl, Microsoft Global Catalog over <u>SSL</u> (similar to port 3268, <u>LDAP over SSL</u> )             |
| 3283      | Yes        | Yes        |      |      | <i>Net Assistant</i> , <sup>[11]</sup> a predecessor to <i>Apple Remote Desktop</i>                             |
|           | Unofficial | Unofficial |      |      | <u>Apple Remote Desktop 2.0 or later</u> <sup>[11]</sup>                                                        |
| 3290      |            | Unofficial |      |      | Virtual Air Traffic Simulation (VATSIM) network voice communication                                             |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                          |
|------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------|
| 3305 | Yes        | Yes        |      |      | <u>Odette File Transfer Protocol (OFTP)</u>                                                                          |
| 3306 | Yes        | Assigned   |      |      | <u>MySQL</u> database system <sup>[11]</sup>                                                                         |
| 3323 | Unofficial | Unofficial |      |      | <u>DECE</u> GEODI Server                                                                                             |
| 3332 |            | Unofficial |      |      | Thundercloud DataPath Overlay Control                                                                                |
| 3333 | Unofficial |            |      |      | <u>Eggdrop</u> , an IRC bot default port <sup>[185]</sup>                                                            |
|      | Unofficial |            |      |      | <u>Network Caller ID</u> server                                                                                      |
|      | Unofficial |            |      |      | <u>CruiseControl.rb</u> <sup>[186]</sup>                                                                             |
|      | Unofficial |            |      |      | <u>OpenOCD (gdbserver)</u> <sup>[187]</sup>                                                                          |
| 3344 | Unofficial | Unofficial |      |      | Repetier-Server                                                                                                      |
| 3351 | Unofficial |            |      |      | <u>Pervasive PSQL</u> <sup>[139]</sup>                                                                               |
| 3386 | Yes        | Yes        |      |      | <u>GTP'</u> <u>3GPP</u> <u>GSM/UMTS</u> <u>CDR</u> logging protocol                                                  |
| 3389 | Yes        | Yes        |      |      | <u>Microsoft Terminal Server (RDP)</u> officially registered as <u>Windows Based Terminal (WBT)</u> <sup>[188]</sup> |
| 3396 | Yes        | Yes        |      |      | <u>Novell NDPS</u> Printer Agent                                                                                     |
| 3412 | Yes        | Yes        |      |      | <u>xmlBlaster</u>                                                                                                    |
| 3423 | Yes        |            |      |      | <u>Xware xTrm</u> Communication Protocol                                                                             |
| 3424 | Yes        |            |      |      | <u>Xware xTrm</u> Communication Protocol over SSL                                                                    |
| 3435 | Yes        | Yes        |      |      | <u>Pacom</u> Security User Port                                                                                      |
| 3455 | Yes        | Yes        |      |      | <u>Resource Reservation Protocol (RSVP)</u>                                                                          |
| 3478 | Yes        | Yes        |      |      | <u>STUN</u> , a protocol for NAT traversal <sup>[189]</sup>                                                          |
|      | Yes        | Yes        |      |      | <u>TURN</u> , a protocol for NAT traversal <sup>[190]</sup> (extension to STUN)                                      |
|      | Yes        | Yes        |      |      | <u>STUN Behavior Discovery</u> . <sup>[191]</sup> See also port 5349.                                                |
| 3479 | Unofficial | Unofficial |      |      | <u>PlayStation Network</u> <sup>[192]</sup>                                                                          |
| 3480 | Unofficial | Unofficial |      |      | <u>PlayStation Network</u> <sup>[192]</sup>                                                                          |
| 3483 |            | Yes        |      |      | <u>Slim Devices</u> discovery protocol                                                                               |
|      | Yes        |            |      |      | <u>Slim Devices SlimProto</u> protocol                                                                               |
| 3493 | Yes        | Yes        |      |      | <u>Network UPS Tools (NUT)</u>                                                                                       |
| 3503 | Yes        | Yes        |      |      | <u>MPLS LSP-echo</u> Port                                                                                            |
| 3516 | Yes        | Yes        |      |      | <u>Smartcard</u> Port                                                                                                |
| 3527 |            | Yes        |      |      | <u>Microsoft Message Queuing</u>                                                                                     |
| 3535 | Unofficial |            |      |      | <u>SMTP</u> alternate <sup>[193]</sup>                                                                               |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                         |
|------|------------|------------|------|------|---------------------------------------------------------------------------------------------------------------------|
| 3544 |            | Yes        |      |      | <u>Teredo tunneling</u>                                                                                             |
| 3551 | Yes        | Yes        |      |      | <u>Apcupsd Information Port</u> <sup>[194]</sup>                                                                    |
| 3601 | Yes        |            |      |      | <u>SAP Message Server Port</u> <sup>[195]</sup>                                                                     |
| 3632 | Yes        | Assigned   |      |      | <u>Distcc</u> , distributed compiler <sup>[11]</sup>                                                                |
| 3645 | Yes        | Yes        |      |      | <u>Cyc</u>                                                                                                          |
| 3655 | Yes        | Yes        |      |      | <u>Advanced Systems Concepts, Inc. ActiveBatch Exec Agent</u> <sup>[196]</sup>                                      |
| 3659 | Yes        | Yes        |      |      | <u>Apple SASL</u> , used by <u>macOS Server Password Server</u> <sup>[11]</sup>                                     |
|      |            | Unofficial |      |      | <u>Battlefield 4</u>                                                                                                |
| 3667 | Yes        | Yes        |      |      | <u>Information Exchange</u>                                                                                         |
| 3671 | Yes        | Yes        |      |      | <u>KNXnet/IP</u> ( <u>EIBnet/IP</u> )                                                                               |
| 3689 | Yes        | Assigned   |      |      | <u>Digital Audio Access Protocol (DAAP)</u> , used by <u>Apple's iTunes</u> and <u>AirPlay</u> <sup>[11]</sup>      |
| 3690 | Yes        | Yes        |      |      | <u>Subversion (SVN)</u> <sup>[11]</sup> version control system                                                      |
| 3702 | Yes        | Yes        |      |      | <u>Web Services Dynamic Discovery (WS-Discovery)</u> , used by various components of <u>Windows Vista</u> and later |
| 3724 | Yes        | Yes        |      |      | Some <u>Blizzard</u> games <sup>[124]</sup>                                                                         |
|      | Unofficial |            |      |      | <u>Club Penguin</u> Disney online game for kids                                                                     |
| 3725 | Yes        | Yes        |      |      | <u>Netia NA-ER Port</u>                                                                                             |
| 3749 | Yes        | Yes        |      |      | <u>CimTrak</u> ( <a href="https://www.cimcor.com/cimtrak/">https://www.cimcor.com/cimtrak/</a> ) registered port    |
| 3768 | Yes        | Yes        |      |      | <u>RBLcheckd</u> server daemon                                                                                      |
| 3784 |            | Yes        |      |      | <u>Bidirectional Forwarding Detection (BFD)</u> for IPv4 and IPv6 (Single Hop) (RFC 5881)                           |
| 3785 |            | Unofficial |      |      | <u>VoIP</u> program used by <u>Ventrilo</u>                                                                         |
| 3799 |            | Yes        |      |      | <u>RADIUS</u> change of authorization                                                                               |
| 3804 | Yes        | Yes        |      |      | <u>Harman Professional HiQnet</u> protocol                                                                          |
| 3825 | Unofficial |            |      |      | <u>RedSeal Networks</u> client/server connection                                                                    |
| 3826 | Yes        | Yes        |      |      | <u>WarMUX</u> game server                                                                                           |
|      | Unofficial |            |      |      | <u>RedSeal Networks</u> client/server connection                                                                    |
| 3835 | Unofficial |            |      |      | <u>RedSeal Networks</u> client/server connection                                                                    |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                             |
|------|------------|------------|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3830 | Yes        | Yes        |      |      | System Management Agent, developed and used by Cerner to monitor and manage solutions                                                                   |
| 3856 | Unofficial | Unofficial |      |      | ERP Server Application used by F10 Software                                                                                                             |
| 3880 | Yes        | Yes        |      |      | IGRS                                                                                                                                                    |
| 3868 | Yes        |            | Yes  |      | <u>Diameter</u> base protocol (RFC 3588)                                                                                                                |
| 3872 | Yes        |            |      |      | <u>Oracle Enterprise Manager</u> Remote Agent                                                                                                           |
| 3900 | Yes        |            |      |      | udt_os, <u>IBM UniData</u> UDT OS <sup>[197]</sup>                                                                                                      |
| 3960 |            | Unofficial |      |      | <u>Warframe</u> online interaction                                                                                                                      |
| 3962 |            | Unofficial |      |      | <u>Warframe</u> online interaction                                                                                                                      |
| 3978 | Unofficial | Unofficial |      |      | <u>OpenTTD</u> game (masterserver and content service)                                                                                                  |
| 3978 | Unofficial |            |      |      | Palo Alto Networks' Panorama management of firewalls and log collectors & pre-PAN-OS 8.0 Panorama-to-managed devices software updates. <sup>[198]</sup> |
| 3979 | Unofficial | Unofficial |      |      | <u>OpenTTD</u> game                                                                                                                                     |
| 3999 | Yes        | Yes        |      |      | Norman distributed scanning service                                                                                                                     |
| 4000 | Unofficial | Unofficial |      |      | <u>Diablo II</u> game                                                                                                                                   |
| 4001 | Unofficial |            |      |      | <u>Microsoft Ants</u> game                                                                                                                              |
|      | Unofficial |            |      |      | CoreOS <u>etcd</u> client communication                                                                                                                 |
| 4018 | Yes        | Yes        |      |      | Protocol information and warnings                                                                                                                       |
| 4035 | Unofficial |            |      |      | IBM Rational Developer for System z Remote System Explorer Daemon                                                                                       |
| 4045 | Unofficial | Unofficial |      |      | <u>Solaris</u> lockd NFS lock daemon/manager                                                                                                            |
| 4050 | Unofficial |            |      |      | Mud Master Chat protocol (MMCP) - Peer-to-peer communications between <u>MUD</u> clients. <sup>[199]</sup>                                              |
| 4069 |            | Yes        |      |      | <u>Minger Email Address Verification Protocol</u> <sup>[200]</sup>                                                                                      |
| 4070 | Unofficial | Unofficial |      |      | Amazon Echo Dot (Amazon Alexa) streaming connection with <u>Spotify</u> <sup>[201]</sup>                                                                |
| 4089 | Yes        | Yes        |      |      | OpenCORE Remote Control Service                                                                                                                         |
| 4090 | Yes        | Yes        |      |      | <u>Kerio</u>                                                                                                                                            |
| 4093 | Yes        | Yes        |      |      | PxPlus Client server interface <u>ProvideX</u>                                                                                                          |
| 4096 | Yes        | Yes        |      |      | <u>Ascom Timeplex</u> Bridge Relay Element (BRE)                                                                                                        |
| 4105 | Yes        | Yes        |      |      | Shofar (ShofarNexus)                                                                                                                                    |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                               |
|-----------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4111      | Yes        | Assigned   |      |      | <a href="#">Xgrid</a> <sup>[11]</sup>                                                                                                                     |
| 4116      | Yes        | Yes        |      |      | Smartcard-TLS                                                                                                                                             |
| 4125      | Unofficial |            |      |      | Microsoft Remote Web Workplace administration                                                                                                             |
| 4172      | Yes        | Yes        |      |      | <a href="#">Teradici PCoIP</a>                                                                                                                            |
| 4190      | Yes        |            |      |      | <a href="#">ManageSieve</a> <sup>[202]</sup>                                                                                                              |
| 4195      | Yes        | Yes        | Yes  | Yes  | <a href="#">AWS</a> protocol for cloud remoting solution                                                                                                  |
| 4197      | Yes        | Yes        |      |      | Harman International's HControl protocol for control and monitoring of Audio, Video, Lighting and Control equipment                                       |
| 4198      | Unofficial | Unofficial |      |      | Couch Potato Android app <sup>[203]</sup>                                                                                                                 |
| 4200      | Unofficial |            |      |      | <a href="#">Angular</a> app                                                                                                                               |
| 4201      | Unofficial |            |      |      | <a href="#">TinyMUD</a> and various derivatives                                                                                                           |
| 4222      | Unofficial |            |      |      | NATS server default port <sup>[204]</sup>                                                                                                                 |
| 4226      | Unofficial | Unofficial |      |      | <a href="#">Aleph One</a> , a computer game                                                                                                               |
| 4242      | Unofficial |            |      |      | <a href="#">Orthanc</a> - <a href="#">DICOM</a> server <sup>[205]</sup>                                                                                   |
|           | Unofficial |            |      |      | <a href="#">Quassel</a> distributed IRC client                                                                                                            |
| 4243      | Unofficial |            |      |      | <a href="#">Docker</a> implementations, redistributions, and setups default <sup>[206]</sup>                                                              |
|           | Unofficial |            |      |      | <a href="#">CrashPlan</a>                                                                                                                                 |
| 4244      | Unofficial | Unofficial |      |      | <a href="#">Viber</a> <sup>[207]</sup>                                                                                                                    |
| 4303      | Yes        | Yes        |      |      | Simple Railroad Command Protocol (SRCP)                                                                                                                   |
| 4307      | Yes        |            |      |      | TrueConf Client - TrueConf Server media data exchange <sup>[208]</sup>                                                                                    |
| 4321      | Yes        |            |      |      | <a href="#">Referral Whois (RWhois) Protocol</a> <sup>[209]</sup>                                                                                         |
| 4444      | Unofficial | Unofficial |      |      | <a href="#">Oracle WebCenter Content</a> : Content Server—Intradoc Socket port. (formerly known as Oracle <a href="#">Universal Content Management</a> ). |
|           | Unofficial |            |      |      | <a href="#">Metasploit</a> 's default listener port <sup>[210]</sup>                                                                                      |
|           | Unofficial | Unofficial |      |      | <a href="#">Xvfb</a> X server virtual frame buffer service                                                                                                |
|           | Unofficial |            |      |      | <a href="#">OpenOCD (Telnet)</a> <sup>[187]</sup>                                                                                                         |
| 4444–4445 | Unofficial |            |      |      | <a href="#">I2P</a> HTTP/S proxy                                                                                                                          |
| 4486      | Yes        | Yes        |      |      | Integrated Client Message Service (ICMS)                                                                                                                  |
| 4488      | Yes        | Assigned   |      |      | Apple Wide Area Connectivity Service, used by <a href="#">Back to My Mac</a> <sup>[11]</sup>                                                              |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                  |
|-----------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 4500      | Assigned   | Yes        |      |      | IPSec NAT Traversal <sup>[11]</sup> (RFC 3947, RFC 4306)                                                                                     |
| 4502–4534 | Yes        |            |      |      | Microsoft Silverlight connectable ports under non-elevated trust                                                                             |
| 4505–4506 | Unofficial |            |      |      | <u>Salt</u> master                                                                                                                           |
| 4534      |            | Unofficial |      |      | <u>Armagetron Advanced</u> server default                                                                                                    |
| 4560      | Unofficial |            |      |      | default <u>Log4j</u> socketappender port                                                                                                     |
| 4567      | Unofficial |            |      |      | <u>Sinatra</u> default server port in development mode (HTTP)                                                                                |
| 4569      |            | Yes        |      |      | <u>Inter-Asterisk eXchange</u> (IAx2)                                                                                                        |
| 4604      | Yes        |            |      |      | <u>Identity Registration Protocol</u>                                                                                                        |
| 4605      | Yes        |            |      |      | <u>Direct End to End Secure Chat Protocol</u>                                                                                                |
| 4610–4640 | Unofficial |            |      |      | <u>QualiSystems</u> TestShell Suite Services                                                                                                 |
| 4662      | Yes        | Yes        |      |      | OrbitNet Message Service                                                                                                                     |
|           | Unofficial |            |      |      | Default for older versions of <u>eMule</u> <sup>[211]</sup>                                                                                  |
| 4664      | Unofficial |            |      |      | <u>Google Desktop Search</u>                                                                                                                 |
| 4672      |            | Unofficial |      |      | Default for older versions of <u>eMule</u> <sup>[211]</sup>                                                                                  |
| 4711      | Unofficial |            |      |      | <u>eMule</u> optional web interface <sup>[211]</sup>                                                                                         |
| 4713      | Unofficial |            |      |      | <u>PulseAudio</u> sound server                                                                                                               |
| 4723      | Unofficial |            |      |      | <u>Appium</u> open source automation tool                                                                                                    |
| 4724      | Unofficial |            |      |      | Default bootstrap port to use on device to talk to <u>Appium</u>                                                                             |
| 4728      | Yes        |            |      |      | Computer Associates Desktop and Server Management (DMP)/Port Multiplexer <sup>[212]</sup>                                                    |
| 4730      | Yes        | Yes        |      |      | <u>Gearman's</u> job server                                                                                                                  |
| 4739      | Yes        | Yes        |      |      | <u>IP Flow Information Export</u>                                                                                                            |
| 4747      | Unofficial |            |      |      | <u>Apprentice</u>                                                                                                                            |
| 4753      | Yes        | Yes        |      |      | SIMON (service and discovery)                                                                                                                |
| 4757      | Unofficial | Unofficial |      |      | <u>Select Studios</u> ( <a href="https://www.select-studios.com">https://www.select-studios.com</a> ) Official Servers                       |
| 4789      |            | Yes        |      |      | Virtual eXtensible Local Area Network ( <u>VXLAN</u> )                                                                                       |
| 4791      |            | Yes        |      |      | <u>IP Routable RoCE</u> (RoCEv2)                                                                                                             |
| 4840      | Yes        | Yes        |      |      | OPC UA Connection Protocol (TCP) and OPC UA Multicast Datagram Protocol (UDP) for <u>OPC Unified Architecture</u> from <u>OPC Foundation</u> |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                        |
|-----------|------------|------------|------|------|--------------------------------------------------------------------------------------------------------------------|
| 4843      | Yes        | Yes        |      |      | OPC UA TCP Protocol over TLS/SSL for OPC Unified Architecture from <a href="#">OPC Foundation</a>                  |
| 4847      | Yes        | Yes        |      |      | Web Fresh Communication, Quadtrion Software & Odorless Entertainment                                               |
| 4848      | Unofficial |            |      |      | Java, Glassfish Application Server administration default                                                          |
| 4894      | Yes        | Yes        |      |      | <a href="#">LysKOM</a> Protocol A                                                                                  |
| 4944      | No         | Unofficial |      |      | <a href="#">DrayTek</a> DSL Status Monitoring <sup>[213]</sup>                                                     |
| 4949      | Yes        |            |      |      | Munin Resource Monitoring Tool                                                                                     |
| 4950      | Yes        | Yes        |      |      | Cylon Controls UC32 Communications Port                                                                            |
| 5000      | Unofficial |            |      |      | UPnP—Windows network device interoperability                                                                       |
|           | Unofficial | Unofficial |      |      | <a href="#">VTun</a> , <a href="#">VPN</a> Software                                                                |
|           | Unofficial |            |      |      | <a href="#">ASP.NET Core</a> — Development Webserver                                                               |
|           |            | Unofficial |      |      | <a href="#">FlightGear</a> multiplayer <sup>[214]</sup>                                                            |
|           | Unofficial |            |      |      | Synology Inc. Management Console, File Station, Audio Station                                                      |
|           | Unofficial |            |      |      | <a href="#">Flask</a> Development Webserver                                                                        |
|           | Unofficial |            |      |      | <a href="#">Heroku</a> console access                                                                              |
|           | Unofficial |            |      |      | <a href="#">Docker Registry</a> <sup>[215]</sup>                                                                   |
|           | Unofficial |            |      |      | AT&T U-verse public, educational, and government access (PEG) streaming over <a href="#">HTTP</a> <sup>[216]</sup> |
|           | Unofficial |            |      |      | High-Speed SECS Message Services                                                                                   |
|           | Unofficial |            |      |      | 3CX Phone System Management Console/Web Client (HTTP)                                                              |
|           | Unofficial |            |      |      | RidgeRun GStreamer Daemon (GSTD) <sup>[217]</sup>                                                                  |
|           | Unofficial |            |      |      | <a href="#">Apple's AirPlay</a> Receiver <sup>[218]</sup>                                                          |
| 5000–5500 | No         | Unofficial |      |      | <i>League of Legends</i> , a multiplayer online battle arena video game <sup>[219]</sup>                           |
| 5001      | Unofficial |            |      |      | <a href="#">Slingbox</a> and Slingplayer                                                                           |
|           | Unofficial | Unofficial |      |      | Iperf (Tool for measuring TCP and UDP bandwidth performance)                                                       |
|           | Unofficial |            |      |      | <a href="#">Synology Inc.</a> Secured Management Console, File Station, Audio Station                              |
|           | Unofficial |            |      |      | 3CX Phone System Management Console/Web Client (HTTPS)                                                             |

| Port | TCP                  | UDP        | SCTP | DCCP | Description                                                                                                                                                          |
|------|----------------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5002 | Unofficial           |            |      |      | ASSA ARX access control system <sup>[220]</sup>                                                                                                                      |
| 5003 | Yes                  | Assigned   |      |      | FileMaker – name binding and transport <sup>[11]</sup>                                                                                                               |
| 5004 | Yes                  | Yes        |      | Yes  | Real-time Transport Protocol media data (RTP) (RFC 3551, RFC 4571)                                                                                                   |
| 5005 | Yes                  | Yes        |      | Yes  | Real-time Transport Protocol control protocol (RTCP) (RFC 3551, RFC 4571)                                                                                            |
| 5007 | Unofficial           |            |      |      | Palo Alto Networks - User-ID agent                                                                                                                                   |
| 5010 | Yes                  | Yes        |      |      | Registered to: TelePath (the IBM FlowMark workflow-management system messaging platform) <sup>[221]</sup><br>The TCP port is now used for: IBM WebSphere MQ Workflow |
| 5011 | Yes                  | Yes        |      |      | TelePath (the IBM FlowMark workflow-management system messaging platform) <sup>[221]</sup>                                                                           |
| 5022 | Unofficial           |            |      |      | MSSQL Server Replication and Database mirroring endpoints <sup>[222]</sup>                                                                                           |
| 5025 | Yes                  | Yes        |      |      | scpi-raw Standard Commands for Programmable Instruments                                                                                                              |
| 5029 |                      | Unofficial |      |      | Sonic Robo Blast 2 and Sonic Robo Blast 2 Kart servers                                                                                                               |
| 5031 | Unofficial           | Unofficial |      |      | AVM CAPI-over-TCP (ISDN over Ethernet tunneling)                                                                                                                     |
| 5037 | Unofficial           |            |      |      | Android ADB server                                                                                                                                                   |
| 5044 | Yes                  |            |      |      | Standard port in Filebeats/Logstash implementation of Lumberjack protocol.                                                                                           |
| 5048 | Yes                  |            |      |      | Texai Message Service                                                                                                                                                |
| 5050 | Unofficial           |            |      |      | Yahoo! Messenger                                                                                                                                                     |
| 5051 | Yes                  |            |      |      | ita-agent Symantec Intruder Alert <sup>[223]</sup>                                                                                                                   |
| 5060 | Yes                  | Yes        |      |      | Session Initiation Protocol (SIP) <sup>[11]</sup>                                                                                                                    |
| 5061 | Yes <sup>[224]</sup> |            |      |      | Session Initiation Protocol (SIP) over TLS                                                                                                                           |
| 5062 | Yes                  | Yes        |      |      | Localisation access                                                                                                                                                  |
| 5064 | Yes                  | Yes        |      |      | EPICS Channel Access server <sup>[225]</sup>                                                                                                                         |
| 5065 | Assigned             | Yes        |      |      | EPICS Channel Access repeater beacon <sup>[225]</sup>                                                                                                                |
| 5070 | Unofficial           | No         |      |      | Binary Floor Control Protocol (BFCP) <sup>[226]</sup>                                                                                                                |
| 5080 | Unofficial           | Unofficial |      |      | List of telephone switches#NEC[NEC Phone System] NEC SV8100 and SV9100 MLC Phones Default iSIP Port                                                                  |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                      |
|------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------|
| 5084 | Yes        | Yes        |      |      | <a href="#">EPCglobal</a> Low Level Reader Protocol ( <a href="#">LLRP</a> )                                                     |
| 5085 | Yes        | Yes        |      |      | <a href="#">EPCglobal</a> Low Level Reader Protocol ( <a href="#">LLRP</a> ) over <a href="#">TLS</a>                            |
| 5090 | Unofficial | Unofficial |      |      | 3CX Phone System 3CX Tunnel Protocol, 3CX App API, 3CX Session Border Controller                                                 |
| 5093 |            | Yes        |      |      | <a href="#">SafeNet, Inc</a> Sentinel LM, Sentinel RMS, License Manager, client-to-server                                        |
| 5099 | Yes        | Yes        |      |      | <a href="#">SafeNet, Inc</a> Sentinel LM, Sentinel RMS, License Manager, server-to-server                                        |
| 5104 | Unofficial |            |      |      | IBM Tivoli Framework NetCOOL/Impact <sup>[227]</sup> <a href="#">HTTP</a> Service                                                |
| 5121 | Unofficial |            |      |      | <a href="#">Neverwinter Nights</a>                                                                                               |
| 5124 | Unofficial | Unofficial |      |      | <a href="#">TorgaNET</a> ( <a href="#">Micronational Darknet</a> )                                                               |
| 5125 | Unofficial | Unofficial |      |      | <a href="#">TorgaNET</a> ( <a href="#">Micronational Intelligence Darknet</a> )                                                  |
| 5150 | Yes        | Yes        |      |      | ATMP Ascend Tunnel Management Protocol <sup>[228]</sup>                                                                          |
| 5151 | Yes        |            |      |      | <a href="#">ESRI</a> SDE Instance                                                                                                |
|      |            | Yes        |      |      | <a href="#">ESRI</a> SDE Remote Start                                                                                            |
| 5154 | Yes        | Yes        |      |      | <a href="#">BZFlag</a>                                                                                                           |
| 5172 | Yes        |            |      |      | PC over IP Endpoint Management <sup>[229]</sup>                                                                                  |
| 5173 | Unofficial |            |      |      | <a href="#">Vite</a> ( <a href="https://vitejs.dev/">https://vitejs.dev/</a> )                                                   |
| 5190 | Yes        | Yes        |      |      | <a href="#">AOL Instant Messenger</a> protocol. <sup>[11]</sup> The chat app is defunct as of 15 December 2017. <sup>[230]</sup> |
| 5198 |            | Unofficial |      |      | <a href="#">EchoLink</a> VoIP Amateur Radio Software (Voice)                                                                     |
| 5199 |            | Unofficial |      |      | <a href="#">EchoLink</a> VoIP Amateur Radio Software (Voice)                                                                     |
| 5200 | Unofficial |            |      |      | <a href="#">EchoLink</a> VoIP Amateur Radio Software (Information)                                                               |
| 5201 | Unofficial | Unofficial |      |      | <a href="#">Iperf3</a> (Tool for measuring TCP and UDP bandwidth performance)                                                    |
| 5222 | Yes        | Reserved   |      |      | Extensible Messaging and Presence Protocol (XMPP) client connection <sup>[11]</sup> <sup>[231]</sup> <sup>[232]</sup>            |
| 5223 | Unofficial |            |      |      | <a href="#">Apple Push Notification Service</a> <sup>[11]</sup> <sup>[157]</sup>                                                 |
|      | Unofficial |            |      |      | Extensible Messaging and Presence Protocol (XMPP) client connection over                                                         |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                                                                               |
|-----------|------------|------------|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |            |            |      |      | <a href="#">SSL</a>                                                                                                                                                                                                                                                       |
| 5228      | Yes        |            |      |      | HP Virtual Room Service                                                                                                                                                                                                                                                   |
|           | Unofficial |            |      |      | <a href="#">Google Play</a> , <a href="#">Android Cloud to Device Messaging Service</a> , <a href="#">Google Cloud Messaging</a>                                                                                                                                          |
| 5235-5236 | Unofficial |            |      |      | <a href="#">Firebase Cloud Messaging</a> <sup>[233]</sup>                                                                                                                                                                                                                 |
| 5242      | Unofficial | Unofficial |      |      | <a href="#">Viber</a> <sup>[207]</sup>                                                                                                                                                                                                                                    |
| 5243      | Unofficial | Unofficial |      |      | <a href="#">Viber</a> <sup>[207]</sup>                                                                                                                                                                                                                                    |
| 5246      |            | Yes        |      |      | Control And Provisioning of Wireless Access Points ( <a href="#">CAPWAP</a> ) CAPWAP control <sup>[234]</sup>                                                                                                                                                             |
| 5247      |            | Yes        |      |      | Control And Provisioning of Wireless Access Points ( <a href="#">CAPWAP</a> ) CAPWAP data <sup>[234]</sup>                                                                                                                                                                |
| 5269      | Yes        |            |      |      | Extensible Messaging and Presence Protocol (XMPP) server-to-server connection <sup>[11][231][232]</sup>                                                                                                                                                                   |
| 5280      | Yes        |            |      |      | Extensible Messaging and Presence Protocol (XMPP) <sup>[235]</sup>                                                                                                                                                                                                        |
| 5281      | Unofficial |            |      |      | Extensible Messaging and Presence Protocol (XMPP) <sup>[236]</sup>                                                                                                                                                                                                        |
| 5298      | Yes        | Yes        |      |      | Extensible Messaging and Presence Protocol (XMPP) <sup>[237]</sup>                                                                                                                                                                                                        |
| 5310      | Assigned   | Yes        |      |      | <i>Outlaws</i> , a 1997 first-person shooter video game <sup>[238]</sup>                                                                                                                                                                                                  |
| 5318      | Yes        | Reserved   |      |      | <a href="#">Certificate Management over CMS</a> <sup>[239]</sup>                                                                                                                                                                                                          |
| 5349      | Yes        | Yes        |      |      | <a href="#">STUN over TLS/DTLS</a> , a protocol for <a href="#">NAT traversal</a> <sup>[189]</sup>                                                                                                                                                                        |
|           | Yes        | Yes        |      |      | <a href="#">TURN over TLS/DTLS</a> , a protocol for <a href="#">NAT traversal</a> <sup>[190]</sup>                                                                                                                                                                        |
|           | Yes        | Reserved   |      |      | <a href="#">STUN Behavior Discovery over TLS</a> . <sup>[191]</sup> See also port 3478.                                                                                                                                                                                   |
| 5351      | Reserved   | Yes        |      |      | <a href="#">NAT Port Mapping Protocol</a> and <a href="#">Port Control Protocol</a> —client-requested configuration for connections through <a href="#">network address translators</a> and <a href="#">firewalls</a>                                                     |
| 5353      | Assigned   | Yes        |      |      | <a href="#">Multicast DNS (mDNS)</a> <sup>[11]</sup>                                                                                                                                                                                                                      |
| 5355      | Yes        | Yes        |      |      | <a href="#">Link-Local Multicast Name Resolution (LLMNR)</a> , allows <a href="#">hosts</a> to perform <a href="#">name resolution</a> for hosts on the same <a href="#">local link</a> (only provided by <a href="#">Windows Vista</a> and <a href="#">Server 2008</a> ) |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                     |
|------|------------|------------|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5357 | Unofficial | Unofficial |      |      | Web Services for Devices (WSDAPI) (only provided by Windows Vista, Windows 7 and Server 2008)                                                                                   |
| 5358 | Unofficial | Unofficial |      |      | WSDAPI Applications to Use a Secure Channel (only provided by Windows Vista, Windows 7 and Server 2008)                                                                         |
| 5394 |            | Unofficial |      |      | Kega Fusion, a Sega multi-console emulator <sup>[240][241]</sup>                                                                                                                |
| 5402 | Yes        | Yes        |      |      | Multicast File Transfer Protocol (MFTP) <sup>[242]</sup>                                                                                                                        |
| 5405 | Yes        | Yes        |      |      | NetSupport Manager                                                                                                                                                              |
| 5412 | Yes        | Yes        |      |      | IBM Rational Synergy (Telelogic Synergy) (Continuous CM) Message Router                                                                                                         |
| 5413 | Yes        | Yes        |      |      | Wonderware SuiteLink service                                                                                                                                                    |
| 5417 | Yes        | Yes        |      |      | SNS Agent                                                                                                                                                                       |
| 5421 | Yes        | Yes        |      |      | NetSupport Manager                                                                                                                                                              |
| 5432 | Yes        | Assigned   |      |      | PostgreSQL <sup>[11]</sup> database system                                                                                                                                      |
| 5433 | Unofficial |            |      |      | Bouwsoft file/webserver <sup>[243]</sup>                                                                                                                                        |
| 5445 |            | Unofficial |      |      | Cisco Unified Video Advantage                                                                                                                                                   |
| 5450 | Unofficial | Unofficial |      |      | OSIsoft PI Server Client Access <sup>[244]</sup>                                                                                                                                |
| 5457 | Unofficial |            |      |      | OSIsoft PI Asset Framework Client Access <sup>[245]</sup>                                                                                                                       |
| 5458 | Unofficial |            |      |      | OSIsoft PI Notifications Client Access <sup>[246]</sup>                                                                                                                         |
| 5480 | Unofficial |            |      |      | VMware VAMI (Virtual Appliance Management Infrastructure)—used for initial setup of various administration settings on Virtual Appliances designed using the VAMI architecture. |
| 5481 | Unofficial |            |      |      | Schneider Electric's ClearSCADA (SCADA implementation for Windows) — used for client-to-server communication. <sup>[247]</sup>                                                  |
| 5495 | Unofficial |            |      |      | IBM Cognos TM1 Admin server                                                                                                                                                     |
| 5498 | Unofficial |            |      |      | Hotline tracker server connection                                                                                                                                               |
| 5499 |            | Unofficial |      |      | Hotline tracker server discovery                                                                                                                                                |
| 5500 | Unofficial |            |      |      | Hotline control connection                                                                                                                                                      |
|      | Unofficial |            |      |      | VNC Remote Frame Buffer RFB protocol—for incoming listening viewer                                                                                                              |
| 5501 | Unofficial |            |      |      | Hotline file transfer connection                                                                                                                                                |
| 5517 | Unofficial |            |      |      | Setiqueue Proxy server client for SETI@Home project                                                                                                                             |
| 5550 | Unofficial |            |      |      | Hewlett-Packard Data Protector                                                                                                                                                  |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                                                                                         |
|------|------------|------------|------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5554 | Unofficial | Unofficial |      |      | <a href="#">Fastboot</a> default wireless port                                                                                                                                                                                                                                      |
| 5555 | Unofficial | Unofficial |      |      | <a href="#">Oracle WebCenter Content: Inbound Refinery—Intradoc Socket</a> port. (formerly known as <a href="#">Oracle Universal Content Management</a> ). Port though often changed during installation                                                                            |
|      | Unofficial |            |      |      | <a href="#">Freeciv</a> versions up to 2.0, <a href="#">Hewlett-Packard Data Protector</a> , <a href="#">McAfee EndPoint Encryption Database Server</a> , <a href="#">SAP</a> , Default for <a href="#">Microsoft Dynamics CRM 4.0</a> , <a href="#">Softether VPN</a> default port |
| 5556 | Yes        | Yes        |      |      | <a href="#">Freeciv</a> , <a href="#">Oracle WebLogic Server Node Manager</a> <sup>[248]</sup>                                                                                                                                                                                      |
| 5568 | Yes        | Yes        |      |      | <a href="#">Session Data Transport (SDT)</a> , a part of <a href="#">Architecture for Control Networks (ACN)</a> <sup>[249]</sup>                                                                                                                                                   |
| 5601 | Unofficial |            |      |      | <a href="#">Kibana</a> <sup>[250]</sup>                                                                                                                                                                                                                                             |
| 5631 | Yes        |            |      |      | <a href="#">pcANYWHEREdata</a> , <a href="#">Symantec pcAnywhere</a> (version 7.52 and later <sup>[251]</sup> <sup>[252]</sup> ) data                                                                                                                                               |
| 5632 |            | Yes        |      |      | <a href="#">pcANYWHEREstat</a> , <a href="#">Symantec pcAnywhere</a> (version 7.52 and later) status                                                                                                                                                                                |
| 5656 | Unofficial |            |      |      | <a href="#">IBM Lotus Sametime</a> p2p file transfer                                                                                                                                                                                                                                |
| 5666 | Unofficial |            |      |      | <a href="#">NRPE</a> ( <a href="#">Nagios</a> )                                                                                                                                                                                                                                     |
| 5667 | Unofficial |            |      |      | <a href="#">NSCA</a> ( <a href="#">Nagios</a> )                                                                                                                                                                                                                                     |
| 5670 | Yes        |            |      |      | <a href="#">FILEMQ ZeroMQ File Message Queuing Protocol</a>                                                                                                                                                                                                                         |
|      |            | Yes        |      |      | <a href="#">ZRE-DISC ZeroMQ Realtime Exchange Protocol (Discovery)</a>                                                                                                                                                                                                              |
| 5671 | Yes        | Assigned   |      |      | <a href="#">Advanced Message Queuing Protocol (AMQP)</a> <sup>[253]</sup> over <a href="#">TLS</a>                                                                                                                                                                                  |
| 5672 | Yes        | Assigned   | Yes  |      | <a href="#">Advanced Message Queuing Protocol (AMQP)</a> <sup>[253]</sup>                                                                                                                                                                                                           |
| 5678 | Unofficial | No         |      |      | <a href="#">n8n</a> <sup>[254]</sup>                                                                                                                                                                                                                                                |
| 5683 | Yes        | Yes        |      |      | <a href="#">Constrained Application Protocol</a> (CoAP)                                                                                                                                                                                                                             |
| 5684 | Yes        | Yes        |      |      | <a href="#">Constrained Application Protocol Secure</a> (CoAPs)                                                                                                                                                                                                                     |
| 5693 | Unofficial |            |      |      | <a href="#">Nagios Cross Platform Agent (NCPA)</a> <sup>[255]</sup>                                                                                                                                                                                                                 |
| 5701 | Unofficial |            |      |      | <a href="#">Hazelcast</a> default communication port <sup>[256]</sup>                                                                                                                                                                                                               |
| 5718 | Unofficial |            |      |      | <a href="#">Microsoft DPM Data Channel</a> (with the agent coordinator)                                                                                                                                                                                                             |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                  |
|-----------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 5719      | Unofficial |            |      |      | Microsoft DPM Data Channel (with the protection agent)                                                                                       |
| 5722      | Yes        | Yes        |      |      | Microsoft RPC, DFSR (SYSVOL) Replication Service                                                                                             |
| 5723      | Unofficial |            |      |      | System Center Operations Manager <sup>[257]</sup>                                                                                            |
| 5724      | Unofficial |            |      |      | Operations Manager Console                                                                                                                   |
| 5741      | Yes        | Yes        |      |      | IDA Discover Port 1                                                                                                                          |
| 5742      | Yes        | Yes        |      |      | IDA Discover Port 2                                                                                                                          |
| 5800      | Unofficial |            |      |      | VNC Remote Frame Buffer <u>RFB protocol</u> over <u>HTTP</u>                                                                                 |
|           | Unofficial |            |      |      | ProjectWise Server <sup>[258]</sup>                                                                                                          |
| 5900      | Yes        | Yes        |      |      | Remote Frame Buffer protocol (RFB)                                                                                                           |
|           | Unofficial |            |      |      | Virtual Network Computing (VNC) Remote Frame Buffer <u>RFB protocol</u> <sup>[11][259]</sup>                                                 |
| 5905      | Unofficial |            |      |      | Windows service "C:\Program Files\Intel\Intel(R) Online Connect Access\IntelTechnologyAccessService.exe" that listens on 127.0.0.1           |
| 5931      | Yes        | Yes        |      |      | AMMY admin Remote Control                                                                                                                    |
| 5938      | Unofficial | Unofficial |      |      | TeamViewer remote desktop protocol <sup>[260]</sup>                                                                                          |
| 5984      | Yes        | Yes        |      |      | CouchDB database server                                                                                                                      |
| 5985      | Yes        |            |      |      | Windows PowerShell Default psSession Port <sup>[261]</sup> Windows Remote Management Service (WinRM-HTTP) <sup>[262]</sup>                   |
| 5986      | Yes        |            |      |      | Windows PowerShell Default psSession Port <sup>[261]</sup> Windows Remote Management Service (WinRM-HTTPS) <sup>[262]</sup>                  |
| 5988-5989 | Yes        |            |      |      | CIM-XML (DMTF Protocol) <sup>[263]</sup>                                                                                                     |
| 6000-6063 | Yes        | Yes        |      |      | X11—used between an X client and server over the network                                                                                     |
| 6005      | Unofficial |            |      |      | Default for BMC Software Control-M/Server—Socket used for communication between Control-M processes—though often changed during installation |
|           | Unofficial |            |      |      | Default for Camfrog chat & cam client                                                                                                        |
| 6009      | Unofficial |            |      |      | JD Edwards EnterpriseOne ERP system JDENet messaging client listener                                                                         |
| 6050      | Unofficial |            |      |      | Arcserve backup                                                                                                                              |
| 6051      | Unofficial |            |      |      | Arcserve backup                                                                                                                              |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                |
|------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------|
| 6086 | Yes        |            |      |      | Peer Distributed Transfer Protocol (PDTP), FTP like file server in a P2P network                                           |
| 6100 | Unofficial |            |      |      | <u>Vizrt</u> System                                                                                                        |
|      | Unofficial |            |      |      | <u>Ventrilo</u> authentication for version 3                                                                               |
| 6101 | Unofficial |            |      |      | Backup Exec Agent Browser                                                                                                  |
| 6110 | Yes        | Yes        |      |      | softcm, <u>HP Softbench</u> CM                                                                                             |
| 6111 | Yes        | Yes        |      |      | spc, <u>HP Softbench</u> Sub-Process Control                                                                               |
| 6112 | Yes        | Yes        |      |      | dtspcd, execute commands and launch applications remotely                                                                  |
|      | Unofficial | Unofficial |      |      | Blizzard's Battle.net gaming service and some games, <sup>[124]</sup> ArenaNet gaming service, <u>Relic</u> gaming service |
|      | Unofficial |            |      |      | <u>Club Penguin</u> Disney online game for kids                                                                            |
| 6113 | Unofficial |            |      |      | <u>Club Penguin</u> Disney online game for kids, Used by some <u>Blizzard</u> games <sup>[124]</sup>                       |
| 6136 | Unofficial |            |      |      | <u>ObjectDB</u> database server <sup>[264]</sup>                                                                           |
| 6159 | Yes        |            |      |      | ARINC 840 <u>EFB</u> Application Control Interface                                                                         |
| 6160 | Unofficial |            |      |      | <u>Veeam</u> Installer Service                                                                                             |
| 6161 | Unofficial |            |      |      | <u>Veeam</u> vPower NFS Service                                                                                            |
| 6162 | Unofficial |            |      |      | <u>Veeam</u> Data Mover                                                                                                    |
| 6163 | Unofficial |            |      |      | <u>Veeam</u> Hyper-V Integration Service                                                                                   |
| 6164 | Unofficial |            |      |      | <u>Veeam</u> WAN Accelerator                                                                                               |
| 6165 | Unofficial |            |      |      | <u>Veeam</u> WAN Accelerator Data Transfer                                                                                 |
| 6167 | Unofficial |            |      |      | <u>Veeam</u> Log Shipping Service                                                                                          |
| 6170 | Unofficial |            |      |      | <u>Veeam</u> Mount Server                                                                                                  |
| 6200 | Unofficial |            |      |      | Oracle WebCenter Content Portable: Content Server (With Native UI) and Inbound Refinery                                    |
| 6201 | Assigned   |            |      |      | Thermo-Calc Software AB: Management of service nodes in a processing grid for thermodynamic calculations                   |
|      | Unofficial |            |      |      | Oracle WebCenter Content Portable: Admin                                                                                   |
| 6225 | Unofficial |            |      |      | Oracle WebCenter Content Portable: Content Server Web UI                                                                   |
| 6227 | Unofficial |            |      |      | Oracle WebCenter Content Portable: JavaDB                                                                                  |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                          |
|-----------|------------|------------|------|------|------------------------------------------------------------------------------------------------------|
| 6240      | Unofficial |            |      |      | Oracle WebCenter Content Portable: Capture                                                           |
| 6244      | Unofficial | Unofficial |      |      | Oracle WebCenter Content Portable: Content Server—Intradoc Socket port                               |
| 6255      | Unofficial | Unofficial |      |      | Oracle WebCenter Content Portable: Inbound Refinery—Intradoc Socket port                             |
| 6257      |            | Unofficial |      |      | <u>WinMX</u> (see also 6699)                                                                         |
| 6260      | Unofficial | Unofficial |      |      | planet M.U.L.E.                                                                                      |
| 6262      | Unofficial |            |      |      | Sybase <u>Advantage Database Server</u>                                                              |
| 6343      |            | Yes        |      |      | <u>SFlow</u> , sFlow traffic monitoring                                                              |
| 6346      | Yes        |            |      |      | <u>gnutella-svc</u> , <u>gnutella</u> ( <u>FrostWire</u> , <u>Limewire</u> , <u>Shareaza</u> , etc.) |
| 6347      | Yes        |            |      |      | <u>gnutella-rtr</u> , Gnutella alternate                                                             |
| 6350      | Yes        |            |      |      | App Discovery and Access Protocol                                                                    |
| 6379      | Yes        |            |      |      | <u>Redis</u> key-value data store                                                                    |
| 6389      | Unofficial |            |      |      | EMC <u>CLARiiON</u>                                                                                  |
| 6432      | Yes        |            |      |      | <u>PgBouncer</u> —A connection pooler for PostgreSQL                                                 |
| 6436      | Unofficial |            |      |      | <u>Leap Motion</u> <u>Websocket Server TLS</u>                                                       |
| 6437      | Unofficial |            |      |      | Leap Motion <u>Websocket Server</u>                                                                  |
| 6443      | Yes        |            |      |      | <u>Kubernetes</u> API server <sup>[265]</sup>                                                        |
| 6444      | Yes        |            |      |      | <u>Sun Grid Engine</u> Qmaster Service                                                               |
| 6445      | Yes        |            |      |      | Sun Grid Engine Execution Service                                                                    |
| 6454      |            | Unofficial |      |      | <u>Art-Net</u> protocol                                                                              |
| 6463–6472 | Unofficial |            |      |      | <u>Discord</u> RPC <sup>[266]</sup>                                                                  |
| 6464      | Yes        |            |      |      | Port assignment for medical device communication in accordance to <u>IEEE 11073-20701</u>            |
| 6513      | Yes        |            |      |      | <u>NETCONF</u> over <u>TLS</u>                                                                       |
| 6514      | Yes        |            |      |      | Syslog over <u>TLS</u> <sup>[267]</sup>                                                              |
| 6515      | Yes        |            |      |      | <u>Elipse</u> RPC Protocol (REC)                                                                     |
| 6516      | Unofficial |            |      |      | <u>Windows Admin Center</u>                                                                          |
| 6543      | Unofficial |            |      |      | Pylons project#Pyramid Default Pylons Pyramid web service port                                       |
| 6556      | Unofficial |            |      |      | <u>Check MK</u> Agent                                                                                |
| 6566      | Yes        |            |      |      | <u>SANE</u> (Scanner Access Now Easy)—SANE network scanner daemon <sup>[268]</sup>                   |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                     |
|-----------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------|
| 6560–6561 | Unofficial |            |      |      | <a href="#">Speech-Dispatcher</a> daemon                                                                        |
| 6571      | Unofficial |            |      |      | <a href="#">Windows Live FolderShare</a> client                                                                 |
| 6600      | Yes        |            |      |      | Microsoft <a href="#">Hyper-V</a> Live                                                                          |
|           | Unofficial |            |      |      | <a href="#">Music Player Daemon</a> (MPD)                                                                       |
| 6601      | Yes        |            |      |      | Microsoft <a href="#">Forefront Threat Management Gateway</a>                                                   |
| 6602      | Yes        |            |      |      | Microsoft Windows WSS Communication                                                                             |
| 6619      | Yes        |            |      |      | <a href="#">odette-ftps</a> , <a href="#">Odette File Transfer Protocol</a> (OFTP) over <a href="#">TLS/SSL</a> |
| 6622      | Yes        |            |      |      | Multicast FTP                                                                                                   |
| 6653      | Yes        | Assigned   |      |      | <a href="#">OpenFlow</a>                                                                                        |
| 6660–6664 | Unofficial |            |      |      | <a href="#">Internet Relay Chat</a> (IRC)                                                                       |
| 6665–6669 | Yes        |            |      |      | Internet Relay Chat (IRC)                                                                                       |
| 6679      | Yes        |            |      |      | Osorno Automation Protocol (OSAUT)                                                                              |
|           | Unofficial |            |      |      | Internet Relay Chat (IRC) <a href="#">SSL</a> (Secure Internet Relay Chat)—often used                           |
| 6690      | Unofficial |            |      |      | Synology Cloud station                                                                                          |
| 6697      | Yes        |            |      |      | IRC <a href="#">SSL</a> (Secure Internet Relay Chat)—often used                                                 |
| 6699      | Unofficial |            |      |      | <a href="#">WinMX</a> (see also 6257)                                                                           |
| 6715      | Unofficial |            |      |      | AberMUD and derivatives default port                                                                            |
| 6771      |            | Unofficial |      |      | BitTorrent <a href="#">Local Peer Discovery</a>                                                                 |
| 6783–6785 | Unofficial |            |      |      | <a href="#">Splashtop Remote</a> server broadcast                                                               |
| 6801      | Yes        |            |      |      | ACNET Control System Protocol                                                                                   |
| 6881–6887 | Unofficial | Unofficial |      |      | BitTorrent beginning of range of ports used most often                                                          |
| 6888      | Yes        |            |      |      | MUSE                                                                                                            |
|           | Unofficial | Unofficial |      |      | BitTorrent continuation of range of ports used most often                                                       |
| 6889–6890 | Unofficial | Unofficial |      |      | BitTorrent continuation of range of ports used most often                                                       |
| 6891–6900 | Unofficial | Unofficial |      |      | BitTorrent continuation of range of ports used most often                                                       |
| 6891–6900 | Unofficial | Unofficial |      |      | <a href="#">Windows Live Messenger</a> (File transfer)                                                          |
| 6901      | Unofficial | Unofficial |      |      | Windows Live Messenger (Voice)                                                                                  |
|           | Unofficial | Unofficial |      |      | BitTorrent continuation of range of ports used most often                                                       |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                   |
|-----------|------------|------------|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6902–6968 | Unofficial | Unofficial |      |      | BitTorrent continuation of range of ports used most often                                                                                                     |
| 6924      | Yes        |            |      |      | split-ping, ping with RX/TX latency/loss split                                                                                                                |
| 6969      | Yes        |            |      |      | acmsoda                                                                                                                                                       |
|           | Unofficial |            |      |      | <u>BitTorrent tracker</u>                                                                                                                                     |
| 6970–6999 | Unofficial | Unofficial |      |      | BitTorrent end of range of ports used most often                                                                                                              |
|           |            | Unofficial |      |      | <u>QuickTime Streaming Server</u> <sup>[11]</sup>                                                                                                             |
| 6980      |            | Unofficial |      |      | <u>Voicemeeter VBAN network audio protocol</u> <sup>[269]</sup>                                                                                               |
| 7000      | Unofficial |            |      |      | Default for Vuze's built-in <u>HTTPS Bittorrent tracker</u>                                                                                                   |
|           | Unofficial |            |      |      | <u>Avira Server Management Console</u>                                                                                                                        |
| 7001      | Unofficial |            |      |      | Avira Server Management Console                                                                                                                               |
|           | Unofficial |            |      |      | Default for <u>BEA WebLogic Server's HTTP server</u> , though often changed during installation                                                               |
| 7002      | Unofficial |            |      |      | Default for BEA WebLogic Server's <u>HTTPS server</u> , though often changed during installation                                                              |
| 7005      | Unofficial |            |      |      | Default for BMC Software <u>Control-M/Server</u> and <u>Control-M/Agent for Agent-to-Server</u> , though often changed during installation                    |
| 7006      | Unofficial |            |      |      | Default for BMC Software <u>Control-M/Server</u> and <u>Control-M/Agent for Server-to-Agent</u> , though often changed during installation                    |
| 7010      | Unofficial |            |      |      | Default for Cisco AON AMC (AON Management Console) <sup>[270]</sup>                                                                                           |
| 7022      | Unofficial |            |      |      | Database mirroring endpoints <sup>[222]</sup>                                                                                                                 |
| 7023      |            | Yes        |      |      | Bryan Wilcutt T2-NMCS Protocol for SatCom Modems                                                                                                              |
| 7025      | Unofficial |            |      |      | Zimbra <u>LMTP</u> [mailbox]—local mail delivery                                                                                                              |
| 7047      | Unofficial |            |      |      | <u>Zimbra conversion server</u>                                                                                                                               |
| 7070      | Unofficial | Unofficial |      |      | <u>Real Time Streaming Protocol (RTSP)</u> , used by <u>QuickTime Streaming Server</u> . TCP is used by default, UDP is used as an alternate. <sup>[11]</sup> |
| 7077      | Yes        |            |      |      | Development-Network Authentication-Protocol                                                                                                                   |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                              |
|-----------|------------|------------|------|------|------------------------------------------------------------------------------------------|
| 7133      | Unofficial |            |      |      | <a href="#">Enemy Territory: Quake Wars</a>                                              |
| 7144      | Unofficial |            |      |      | <a href="#">Peercast</a>                                                                 |
| 7145      | Unofficial |            |      |      | <a href="#">Peercast</a>                                                                 |
| 7171      | Unofficial |            |      |      | <a href="#">Tibia</a>                                                                    |
| 7262      | Yes        |            |      |      | <a href="#">CNAP</a> (Calypso Network Access Protocol)                                   |
| 7272      | Yes        |            |      |      | <a href="#">WatchMe</a> - WatchMe Monitoring                                             |
| 7306      | Unofficial |            |      |      | <a href="#">Zimbra mysql</a> [mailbox]                                                   |
| 7307      | Unofficial |            |      |      | <a href="#">Zimbra mysql</a> [logger]                                                    |
| 7312      |            | Unofficial |      |      | <a href="#">Sibelius</a> License Server                                                  |
| 7396      | Unofficial |            |      |      | Web control interface for <a href="#">Folding@home</a> v7.3.6 and later <sup>[271]</sup> |
| 7400      | Yes        |            |      |      | <a href="#">RTPS</a> (Real Time Publish Subscribe) <a href="#">DDS</a> Discovery         |
| 7401      | Yes        |            |      |      | <a href="#">RTPS</a> (Real Time Publish Subscribe) <a href="#">DDS</a> User-Traffic      |
| 7402      | Yes        |            |      |      | <a href="#">RTPS</a> (Real Time Publish Subscribe) <a href="#">DDS</a> Meta-Traffic      |
| 7471      | Unofficial |            |      |      | <a href="#">Stateless Transport Tunneling</a> (STT)                                      |
| 7473      | Yes        |            |      |      | <a href="#">Rise: The Vieneo Province</a>                                                |
| 7474      | Yes        |            |      |      | <a href="#">Neo4J</a> Server webadmin <sup>[272]</sup>                                   |
| 7478      | Yes        |            |      |      | Default port used by <a href="#">Open iT Server</a> . <sup>[273]</sup>                   |
| 7542      | Yes        |            |      |      | <a href="#">Saratoga</a> file transfer protocol <sup>[274][275]</sup>                    |
| 7547      | Yes        |            |      |      | <a href="#">CPE WAN Management Protocol</a> (CWMP) Technical Report 069                  |
| 7575      |            | Unofficial |      |      | <a href="#">Populous: The Beginning</a> server                                           |
| 7624      | Yes        |            |      |      | <a href="#">Instrument Neutral Distributed Interface</a>                                 |
| 7631      | Yes        |            |      |      | <a href="#">ERLPhase</a>                                                                 |
| 7634      | Unofficial |            |      |      | <a href="#">hddtemp</a> —Utility to monitor hard drive temperature                       |
| 7652–7654 | Unofficial |            |      |      | <a href="#">I2P</a> anonymizing overlay network                                          |
| 7655      |            | Unofficial |      |      | <a href="#">I2P SAM Bridge</a> Socket API                                                |
| 7656–7660 | Unofficial |            |      |      | <a href="#">I2P</a> anonymizing overlay network                                          |
| 7670      | Unofficial |            |      |      | <a href="#">BrettspielWelt</a> BSW Boardgame Portal                                      |
| 7680      | Unofficial |            |      |      | Delivery Optimization for <a href="#">Windows 10</a> <sup>[276]</sup>                    |
| 7687      | Yes        |            |      |      | <a href="#">Bolt</a> database connection                                                 |
| 7707–7708 |            | Unofficial |      |      | <a href="#">Killing Floor</a>                                                            |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                    |
|-----------|------------|------------|------|------|--------------------------------------------------------------------------------------------------------------------------------|
| 7717      |            | Unofficial |      |      | <i>Killing Floor</i>                                                                                                           |
| 7777      | Unofficial |            |      |      | iChat server file transfer proxy <sup>[11]</sup>                                                                               |
|           | Unofficial |            |      |      | Oracle Cluster File System 2                                                                                                   |
|           | Unofficial |            |      |      | Windows backdoor program tini.exe default                                                                                      |
|           | Unofficial |            |      |      | <i>Just Cause 2: Multiplayer Mod</i> Server                                                                                    |
|           | Unofficial |            |      |      | <i>Terraria</i> default server                                                                                                 |
|           |            | Unofficial |      |      | <i>San Andreas Multiplayer</i> (SA-MP) default port server                                                                     |
|           |            | Unofficial |      |      | <i>SCP: Secret Laboratory</i> Multiplayer Server                                                                               |
| 7777–7788 | Unofficial | Unofficial |      |      | <i>Unreal Tournament</i> series default server                                                                                 |
| 7831      | Unofficial |            |      |      | Default used by Smartlaunch Internet Cafe Administration <sup>[277]</sup> software                                             |
| 7880      | Unofficial | Unofficial |      |      | PowerSchool Gradebook Server                                                                                                   |
| 7890      | Unofficial |            |      |      | Default that will be used by the iControl Internet Cafe Suite Administration software                                          |
| 7915      | Unofficial |            |      |      | Default for YSFlight server <sup>[278]</sup>                                                                                   |
| 7935      | Unofficial |            |      |      | Fixed port used for Adobe Flash Debug Player to communicate with a debugger (Flash IDE, Flex Builder or fdb). <sup>[279]</sup> |
| 7946      | Unofficial | Unofficial |      |      | Docker Swarm communication among nodes <sup>[164]</sup>                                                                        |
| 7979      |            | Unofficial |      |      | Used by SilverBluff Studios for communication between servers and clients.                                                     |
| 7990      | Unofficial |            |      |      | Atlassian <i>Bitbucket</i> (default port)                                                                                      |
| 8000      | Unofficial |            |      |      | Commonly used for Internet radio streams such as <i>SHOUTcast</i> , <i>Icecast</i> and <i>iTunes Radio</i> <sup>[11]</sup>     |
|           | Unofficial |            |      |      | <i>DynamoDB Local</i> <sup>[280]</sup>                                                                                         |
|           | Unofficial |            |      |      | <i>Django</i> Development Webserver <sup>[281]</sup>                                                                           |
|           | Unofficial |            |      |      | <i>Python 3</i> http.server <sup>[282]</sup>                                                                                   |
| 8005      | Unofficial |            |      |      | <i>Tomcat</i> remote shutdown <sup>[11]</sup>                                                                                  |
|           | Unofficial |            |      |      | <i>PLATO</i> ASCII protocol (RFC 600)                                                                                          |
|           | Unofficial |            |      |      | Windows SCCM HTTP listener service <sup>[283]</sup>                                                                            |
| 8006      | Unofficial |            |      |      | <i>Quest AppAssure 5 API</i> <sup>[284]</sup>                                                                                  |
|           | Unofficial | No         |      |      | <i>Proxmox Virtual Environment</i> admin web interface <sup>[285]</sup>                                                        |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                             |
|------|------------|------------|------|------|-----------------------------------------------------------------------------------------|
| 8007 | Unofficial |            |      |      | Quest AppAssure 5 Engine <sup>[284]</sup>                                               |
| 8007 | Yes        |            |      |      | Proxmox Backup Server admin web interface                                               |
| 8008 | Unofficial | Unofficial |      |      | Alternative port for HTTP. See also ports 80 and 8080.                                  |
|      | Unofficial |            |      |      | IBM HTTP Server administration default                                                  |
|      | Unofficial |            |      |      | iCal, a calendar application by Apple <sup>[11]</sup>                                   |
|      | Unofficial | No         |      |      | Matrix homeserver federation over HTTP <sup>[286]</sup>                                 |
| 8009 | Unofficial |            |      |      | Apache JServ Protocol (ajp13)                                                           |
| 8010 | Unofficial | No         |      |      | Buildbot web status page                                                                |
| 8042 | Unofficial |            |      |      | Orthanc – REST API over HTTP <sup>[205]</sup>                                           |
| 8069 | Unofficial |            |      |      | OpenERP 5.0 XML-RPC protocol <sup>[287]</sup>                                           |
| 8070 | Unofficial |            |      |      | OpenERP 5.0 NET-RPC protocol <sup>[287]</sup>                                           |
| 8074 | Yes        |            |      |      | Gadu-Gadu                                                                               |
| 8075 | Unofficial |            |      |      | <i>Killing Floor</i> web administration interface                                       |
| 8080 | Yes        |            |      |      | Alternative port for HTTP. See also ports 80 and 8008.                                  |
|      | Unofficial |            |      |      | Apache Tomcat <sup>[288]</sup>                                                          |
|      | Unofficial |            |      |      | Atlassian JIRA applications <sup>[289]</sup>                                            |
| 8081 | Yes        | Yes        |      |      | Sun Proxy Admin Service <sup>[290]</sup>                                                |
| 8088 | Unofficial |            |      |      | Asterisk management access via HTTP                                                     |
| 8089 | Unofficial | No         |      |      | Splunk daemon management <sup>[291]</sup>                                               |
|      | Unofficial |            |      |      | Fritz!Box automatic TR-069 configuration <sup>[292]</sup>                               |
| 8090 | Unofficial |            |      |      | Atlassian Confluence <sup>[293]</sup>                                                   |
|      | Unofficial |            |      |      | Coral Content Distribution Network (legacy; 80 and 8080 now supported) <sup>[294]</sup> |
|      | Unofficial |            |      |      | Matrix identity server                                                                  |
| 8091 | Unofficial |            |      |      | CouchBase web administration <sup>[295]</sup>                                           |
| 8092 | Unofficial |            |      |      | CouchBase API <sup>[295]</sup>                                                          |
| 8096 | Unofficial |            |      |      | Emby and Jellyfin HTTP port <sup>[296]</sup>                                            |
| 8111 | Unofficial |            |      |      | JOSM Remote Control                                                                     |
| 8112 | Unofficial |            |      |      | PAC Pacifica Coin                                                                       |
| 8116 |            | Unofficial |      |      | Check Point Cluster Control Protocol                                                    |

| Port      | TCP        | UDP        | SCTP | DCCP | Description                                                                                                           |
|-----------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------|
| 8118      | Yes        |            |      |      | <a href="#">Privoxy</a> —advertisement-filtering Web proxy                                                            |
| 8123      | Unofficial |            |      |      | <a href="#">Polipo</a> Web proxy                                                                                      |
|           | Unofficial |            |      |      | <a href="#">Home Assistant</a> Home automation                                                                        |
|           | Unofficial |            |      |      | BURST P2P <sup>[297]</sup>                                                                                            |
| 8124      | Unofficial |            |      |      | Standard BURST Mining Pool Software Port                                                                              |
| 8125      | Unofficial |            |      |      | BURST Web Interface                                                                                                   |
| 8139      | Unofficial |            |      |      | <a href="#">Puppet (software)</a> Client agent                                                                        |
| 8140      | Yes        |            |      |      | <a href="#">Puppet (software)</a> Master server                                                                       |
| 8172      | Unofficial |            |      |      | <a href="#">Microsoft Remote Administration for IIS Manager</a> <sup>[298]</sup>                                      |
| 8184      | Unofficial |            |      |      | <a href="#">NCSA Brown Dog</a> Data Access Proxy                                                                      |
| 8194–8195 | Yes        |            |      |      | <a href="#">Bloomberg Terminal</a> <sup>[299]</sup>                                                                   |
| 8200      | Unofficial |            |      |      | <a href="#">GoToMyPC</a>                                                                                              |
|           | Unofficial |            |      |      | MiniDLNA media server Web Interface                                                                                   |
| 8222      | Unofficial |            |      |      | <a href="#">VMware VI</a> Web Access via HTTP <sup>[300]</sup>                                                        |
| 8236      | Unofficial |            |      |      | <a href="#">jRCS</a> listener for <a href="#">Rocket Software jBASE</a> Remote Connectivity Server <sup>[301]</sup>   |
| 8243      | Yes        |            |      |      | <a href="#">HTTPS</a> listener for <a href="#">Apache Synapse</a> <sup>[302]</sup>                                    |
| 8245      | Unofficial |            |      |      | <a href="#">Dynamic DNS</a> for at least <a href="#">No-IP</a> and <a href="#">DynDNS</a> <sup>[303]</sup>            |
| 8280      | Yes        |            |      |      | <a href="#">HTTP</a> listener for <a href="#">Apache Synapse</a> <sup>[302]</sup>                                     |
| 8281      | Unofficial |            |      |      | HTTP Listener for Gatecraft Plugin                                                                                    |
| 8291      | Unofficial |            |      |      | Winbox—Default on a MikroTik RouterOS for a Windows application used to administer MikroTik RouterOS <sup>[304]</sup> |
| 8303      |            | Unofficial |      |      | <a href="#">Teeworlds</a> Server                                                                                      |
| 8332      | Unofficial |            |      |      | <a href="#">Bitcoin JSON-RPC</a> server <sup>[305]</sup>                                                              |
| 8333      | Unofficial |            |      |      | <a href="#">Bitcoin</a>                                                                                               |
|           | Unofficial |            |      |      | <a href="#">VMware VI</a> Web Access via HTTPS <sup>[300]</sup>                                                       |
| 8334      | Unofficial |            |      |      | <a href="#">Filestash</a> server (default) <sup>[306]</sup>                                                           |
| 8337      | Unofficial |            |      |      | <a href="#">VisualSVN</a> Distributed File System Service (VDFS) <sup>[307]</sup>                                     |
| 8384      | Unofficial |            |      |      | <a href="#">Syncthing</a> web GUI                                                                                     |
| 8388      | Unofficial |            |      |      | <a href="#">Shadowsocks</a> proxy server                                                                              |

| Port | TCP        | UDP | SCTP | DCCP | Description                                                                                                                                                                                                                         |
|------|------------|-----|------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8400 | Yes        |     |      |      | Commvault Communications Service ( <a href="https://documentation.commvault.com/commvault/v11/article?p=8572.htm">https://documentation.commvault.com/commvault/v11/article?p=8572.htm</a> ) (GxCVD, found in all client computers) |
| 8401 | Yes        |     |      |      | Commvault Server Event Manager (GxEvMgrS, available in CommServe)                                                                                                                                                                   |
| 8403 | Yes        |     |      |      | Commvault Firewall (GxFWD, tunnel port for HTTP/HTTPS)                                                                                                                                                                              |
| 8443 | Unofficial |     |      |      | SW Soft Plesk Control Panel                                                                                                                                                                                                         |
|      | Unofficial |     |      |      | Apache Tomcat SSL                                                                                                                                                                                                                   |
|      | Unofficial |     |      |      | Promise WebPAM SSL                                                                                                                                                                                                                  |
|      | Unofficial |     |      |      | iCal over <a href="#">SSL</a> <sup>[11]</sup>                                                                                                                                                                                       |
|      | Unofficial |     |      |      | MineOs WebUi                                                                                                                                                                                                                        |
| 8444 | Unofficial |     |      |      | Bitmessage                                                                                                                                                                                                                          |
| 8448 | Unofficial | No  |      |      | Matrix homeserver federation over <a href="#">HTTPS</a> <sup>[286]</sup>                                                                                                                                                            |
| 8484 | Unofficial |     |      |      | MapleStory Login Server                                                                                                                                                                                                             |
| 8500 | Unofficial |     |      |      | Adobe ColdFusion built-in web server <sup>[308]</sup>                                                                                                                                                                               |
| 8530 | Unofficial |     |      |      | Windows Server Update Services over <a href="#">HTTP</a> , when using the default role installation settings in Windows Server 2012 and later versions. <sup>[309][310]</sup>                                                       |
| 8531 | Unofficial |     |      |      | Windows Server Update Services over <a href="#">HTTPS</a> , when using the default role installation settings in Windows Server 2012 and later versions. <sup>[309][310]</sup>                                                      |
| 8555 | Unofficial |     |      |      | Symantec DLP OCR Engine <sup>[311]</sup>                                                                                                                                                                                            |
| 8580 | Unofficial |     |      |      | Freegate, an Internet <a href="#">anonymizer</a> and proxy tool <sup>[312]</sup>                                                                                                                                                    |
| 8629 | Unofficial |     |      |      | Tibero database                                                                                                                                                                                                                     |
| 8642 | Unofficial |     |      |      | Lotus Notes Traveler auto synchronization for Windows Mobile and Nokia devices <sup>[313]</sup>                                                                                                                                     |
| 8691 | Unofficial |     |      |      | Ultra Fractal, a <a href="#">fractal generation and rendering software application</a> - distributed calculations over networked computers <sup>[314][315]</sup>                                                                    |
| 8765 | Unofficial | No  |      |      | Default port of a local GUN relay peer that the Internet Archive <sup>[316]</sup> and others use as a decentralized mirror for censorship resistance. <sup>[317]</sup>                                                              |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                         |
|------|------------|------------|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8767 |            | Unofficial |      |      | Voice channel of TeamSpeak 2, <sup>[318]</sup> a proprietary Voice over IP protocol targeted at gamers                                                              |
| 8834 | Unofficial |            |      |      | Nessus, a vulnerability scanner - remote XML-RPC web server <sup>[319]</sup>                                                                                        |
| 8840 | Unofficial |            |      |      | Opera Unite, an extensible framework for web applications <sup>[320][321]</sup>                                                                                     |
| 8880 | Yes        |            |      |      | Alternate port of CDDB (Compact Disc Database) protocol, used to look up audio CD (compact disc) information over the Internet. <sup>[322]</sup> See also port 888. |
|      | Unofficial |            |      |      | IBM WebSphere Application Server SOAP connector <sup>[323]</sup>                                                                                                    |
| 8883 | Yes        |            |      |      | Secure MQTT (MQTT over TLS) <sup>[324][325]</sup>                                                                                                                   |
| 8887 | Unofficial |            |      |      | HyperVM over HTTP                                                                                                                                                   |
| 8888 | Unofficial |            |      |      | HyperVM over HTTPS                                                                                                                                                  |
|      | Unofficial | No         |      |      | Freenet web UI (localhost only)                                                                                                                                     |
|      | Unofficial |            |      |      | Default for IPython <sup>[326]</sup> / Jupyter <sup>[327]</sup> notebook dashboards                                                                                 |
|      | Unofficial |            |      |      | MAMP <sup>[328]</sup>                                                                                                                                               |
| 8889 | Unofficial |            |      |      | MAMP <sup>[328]</sup>                                                                                                                                               |
| 8920 | Unofficial |            |      |      | Jellyfin HTTPS port <sup>[296]</sup>                                                                                                                                |
| 8983 | Unofficial |            |      |      | Apache Solr <sup>[329]</sup>                                                                                                                                        |
| 8997 | Unofficial |            |      |      | Alternate port for I2P Monotone Proxy <sup>[175]</sup>                                                                                                              |
| 8998 | Unofficial |            |      |      | I2P Monotone Proxy <sup>[175]</sup>                                                                                                                                 |
| 8999 | Unofficial |            |      |      | Alternate port for I2P Monotone Proxy <sup>[175]</sup>                                                                                                              |
| 9000 | Unofficial |            |      |      | SonarQube Web Server <sup>[330]</sup>                                                                                                                               |
|      | Unofficial |            |      |      | ClickHouse default port                                                                                                                                             |
|      | Unofficial |            |      |      | DBGp                                                                                                                                                                |
|      | Unofficial |            |      |      | SqueezeCenter web server & streaming                                                                                                                                |
|      |            | Unofficial |      |      | UDPCast                                                                                                                                                             |
|      | Unofficial |            |      |      | Play Framework web server <sup>[331]</sup>                                                                                                                          |
|      | Unofficial |            |      |      | Hadoop NameNode default port                                                                                                                                        |
|      | Unofficial |            |      |      | PHP-FPM default port                                                                                                                                                |
|      | Unofficial |            |      |      | QBittorrent's embedded torrent tracker default port <sup>[332]</sup>                                                                                                |
| 9001 | Yes        |            |      |      | ETL Service Manager <sup>[333]</sup>                                                                                                                                |

| Port      | TCP                     | UDP      | SCTP | DCCP | Description                                                                                                  |
|-----------|-------------------------|----------|------|------|--------------------------------------------------------------------------------------------------------------|
|           | Unofficial              |          |      |      | Microsoft <a href="#">SharePoint</a> authoring environment                                                   |
|           | Unofficial              |          |      |      | cisco-xremote router configuration                                                                           |
|           | Unofficial              |          |      |      | <a href="#">Tor</a> network default                                                                          |
|           | Unofficial              |          |      |      | <a href="#">DBGp</a> Proxy                                                                                   |
|           | Unofficial              |          |      |      | <a href="#">HSQLDB</a> default port                                                                          |
| 9002      | Unofficial              |          |      |      | Newforma Server comms                                                                                        |
| 9006      | Reserved <sup>[2]</sup> |          |      |      |                                                                                                              |
|           | Unofficial              |          |      |      | <a href="#">Tomcat</a> in standalone mode <sup>[11]</sup>                                                    |
| 9030      | Unofficial              |          |      |      | <a href="#">Tor</a> often used                                                                               |
| 9042      | Unofficial              |          |      |      | <a href="#">Apache Cassandra</a> native protocol clients                                                     |
| 9043      | Unofficial              |          |      |      | <a href="#">WebSphere Application Server</a> Administration Console secure                                   |
| 9050–9051 | Unofficial              |          |      |      | <a href="#">Tor</a> (SOCKS-5 proxy client)                                                                   |
| 9060      | Unofficial              |          |      |      | <a href="#">WebSphere Application Server</a> Administration Console                                          |
| 9080      | Yes                     |          |      |      | <a href="#">glrpc</a> , <a href="#">Groove</a> <a href="#">Collaboration software</a> GLRPC                  |
|           | Unofficial              |          |      |      | <a href="#">WebSphere Application Server</a> <a href="#">HTTP Transport</a> (port 1) <a href="#">default</a> |
|           | Unofficial              |          |      |      | Remote Potato by FatAttitude, Windows Media Center addon                                                     |
|           | Unofficial              |          |      |      | ServerWMC, Windows Media Center addon                                                                        |
| 9081      | Unofficial              |          |      |      | <a href="#">Zerto</a> ZVM to ZVM communication                                                               |
| 9090      | Unofficial              |          |      |      | <a href="#">Prometheus</a> metrics server                                                                    |
|           | Unofficial              |          |      |      | <a href="#">Openfire</a> Administration Console                                                              |
|           | Unofficial              |          |      |      | <a href="#">SqueezeCenter</a> control (CLI)                                                                  |
|           | Unofficial              |          |      |      | <a href="#">Cherokee</a> Admin Panel                                                                         |
| 9091      | Unofficial              |          |      |      | <a href="#">Openfire</a> Administration Console (SSL Secured)                                                |
|           | Unofficial              |          |      |      | <a href="#">Transmission (BitTorrent client)</a> Web Interface                                               |
| 9092      | Unofficial              |          |      |      | <a href="#">H2 (DBMS)</a> Database Server                                                                    |
|           | Unofficial              |          |      |      | <a href="#">Apache Kafka</a> A Distributed Streaming Platform <sup>[334]</sup>                               |
| 9100      | Yes                     | Assigned |      |      | <a href="#">PDL</a> Data Stream, used for printing to certain network printers <sup>[11]</sup>               |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                              |
|------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------|
| 9101 | Yes        |            |      |      | <a href="#">Bacula</a> Director                                                                          |
| 9102 | Yes        |            |      |      | <a href="#">Bacula</a> File Daemon                                                                       |
| 9103 | Yes        |            |      |      | <a href="#">Bacula</a> Storage Daemon                                                                    |
| 9119 | Yes        |            |      |      | <a href="#">MXit</a> Instant Messenger                                                                   |
| 9150 | Unofficial |            |      |      | <a href="#">Tor</a> Browser                                                                              |
| 9191 | Unofficial |            |      |      | Sierra Wireless Airlink                                                                                  |
| 9199 | Unofficial |            |      |      | Avtex LLC—qStats                                                                                         |
| 9200 | Unofficial |            |      |      | Elasticsearch <sup>[335]</sup> —default Elasticsearch port                                               |
| 9217 | Unofficial |            |      |      | iPass Platform Service                                                                                   |
| 9293 | Unofficial |            |      |      | Sony PlayStation RemotePlay <sup>[336]</sup>                                                             |
| 9295 | Unofficial | Unofficial |      |      | Sony PlayStation Remote Play Session creation communication port                                         |
| 9296 |            | Unofficial |      |      | Sony PlayStation Remote Play                                                                             |
| 9897 |            | Unofficial |      |      | Sony PlayStation Remote Play Video stream                                                                |
| 9300 | Unofficial |            |      |      | <a href="#">IBM Cognos BI</a>                                                                            |
| 9303 |            | Unofficial |      |      | <a href="#">D-Link Shareport</a> Share storage and MFP printers                                          |
| 9306 | Yes        |            |      |      | <a href="#">Sphinx</a> Native API                                                                        |
| 9309 | Unofficial | Unofficial |      |      | Sony PlayStation Vita Host Collaboration WiFi Data Transfer <sup>[337]</sup>                             |
| 9312 | Yes        |            |      |      | <a href="#">Sphinx</a> SphinxQL                                                                          |
| 9332 | Unofficial |            |      |      | <a href="#">Litecoin</a> JSON-RPC server                                                                 |
| 9333 | Unofficial |            |      |      | <a href="#">Litecoin</a>                                                                                 |
| 9339 | Unofficial |            |      |      | Used by all Supercell games such as Brawl Stars and Clash of Clans, mobile freemium strategy video games |
| 9389 | Yes        |            |      |      | adws, <a href="#">Microsoft AD DS</a> Web Services, <a href="#">Powershell</a> uses this port            |
| 9392 | Unofficial | No         |      |      | OpenVAS Greenbone Security Assistant web interface                                                       |
| 9418 | Yes        |            |      |      | git, <a href="#">Git</a> pack transfer service                                                           |
| 9419 | Unofficial |            |      |      | <a href="#">MooseFS</a> distributed file system - master control port <sup>[338]</sup>                   |
| 9420 | Unofficial |            |      |      | <a href="#">MooseFS</a> distributed file system - master command port <sup>[338]</sup>                   |

| Port | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                    |
|------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 9421 | Unofficial |            |      |      | MooseFS distributed file system - master client port <sup>[338]</sup>                                                                          |
| 9422 | Unofficial |            |      |      | MooseFS distributed file system - Chunkservers <sup>[338]</sup>                                                                                |
| 9425 | Unofficial |            |      |      | MooseFS distributed file system - CGI server <sup>[338]</sup>                                                                                  |
| 9443 | Unofficial |            |      |      | VMware Websense Triton console (HTTPS port used for accessing and administrating a vCenter Server via the Web Management Interface)            |
|      | Unofficial |            |      |      | NCSA Brown Dog Data Tilling Service                                                                                                            |
| 9535 | Yes        |            |      |      | mngsuite, LANDesk Management Suite Remote Control                                                                                              |
| 9536 | Yes        |            |      |      | laes-bf, IP Fabrics Surveillance buffering function                                                                                            |
| 9600 | No         | Unofficial |      |      | Factory Interface Network Service (FINS), a network protocol used by Omron programmable logic controllers                                      |
| 9669 | Unofficial | No         |      |      | VGG Image Search Engine VISE ( <a href="https://www.robots.ox.ac.uk/~vgg/software/vise/">https://www.robots.ox.ac.uk/~vgg/software/vise/</a> ) |
| 9675 | Unofficial | Unofficial |      |      | Spiceworks Desktop, IT Helpdesk Software                                                                                                       |
| 9676 | Unofficial | Unofficial |      |      | Spiceworks Desktop, IT Helpdesk Software                                                                                                       |
| 9695 | Yes        |            |      |      | Content centric networking (CCN, CCNx)                                                                                                         |
| 9735 | Unofficial |            |      |      | Bitcoin Lightning Network <sup>[339]</sup>                                                                                                     |
| 9785 | Unofficial | Unofficial |      |      | Viber <sup>[207]</sup>                                                                                                                         |
| 9800 | Yes        |            |      |      | WebDAV Source                                                                                                                                  |
|      | Unofficial |            |      |      | WebCT e-learning portal                                                                                                                        |
| 9875 | Unofficial |            |      |      | Club Penguin Disney online game for kids                                                                                                       |
| 9898 | Unofficial |            |      |      | Tripwire—File Integrity Monitoring Software <sup>[340]</sup>                                                                                   |
| 9899 |            | Yes        |      |      | SCTP tunneling (port number used in SCTP packets encapsulated in UDP, RFC 6951)                                                                |
| 9901 | Unofficial |            |      |      | Banana for Apache Solr                                                                                                                         |
| 9981 | Unofficial |            |      |      | Tvheadend HTTP server (web interface) <sup>[341]</sup>                                                                                         |
| 9982 | Unofficial |            |      |      | Tvheadend HTSP server (Streaming protocol) <sup>[341]</sup>                                                                                    |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                                                          |
|-------------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9987        | Unofficial | No         |      |      | TeamSpeak 3 server default (voice) port (for the conflicting service see the IANA list) <sup>[342]</sup>                                                                                                                                             |
| 9993        |            | Unofficial |      |      | <u>ZeroTier</u> Default port for ZeroTier                                                                                                                                                                                                            |
| 9997        | Unofficial |            |      |      | Splunk port for communication between the forwarders and indexers                                                                                                                                                                                    |
| 9999        | Unofficial |            |      |      | <u>Urchin</u> Web Analytics                                                                                                                                                                                                                          |
| 9999        | Unofficial |            |      |      | <u>Dash</u> (cryptocurrency) <sup>[343]</sup>                                                                                                                                                                                                        |
| 10000       | Yes        |            |      |      | Network Data Management Protocol (NDMP) Control stream for network backup and restore.                                                                                                                                                               |
|             | Unofficial |            |      |      | <u>BackupExec</u>                                                                                                                                                                                                                                    |
|             | Unofficial |            |      |      | <u>Webmin</u> , Web-based Unix/Linux system administration tool (default port)                                                                                                                                                                       |
| 10000–20000 | No         | Unofficial |      |      | Used on <u>VoIP</u> networks for receiving and transmitting voice telephony traffic which includes <u>Google Voice</u> via the <u>OBiTalk ATA</u> devices as well as on the <u>MagicJack</u> and <u>Vonage</u> ATA network devices. <sup>[344]</sup> |
| 10001       |            | Unofficial |      |      | Ubiquiti UniFi access points broadcast to 255.255.255.255:10001 (UDP) to locate the controller(s)                                                                                                                                                    |
| 10009       | Unofficial | Unofficial |      |      | <u>Crossfire</u> , a multiplayer online First Person Shooter                                                                                                                                                                                         |
| 10011       | Unofficial | No         |      |      | <u>TeamSpeak 3</u> ServerQuery <sup>[342]</sup>                                                                                                                                                                                                      |
| 10022       | Unofficial | No         |      |      | <u>TeamSpeak 3</u> ServerQuery over SSH                                                                                                                                                                                                              |
| 10024       | Unofficial |            |      |      | Zimbra smtp [mta]—to amavis from postfix                                                                                                                                                                                                             |
| 10025       | Unofficial |            |      |      | Zimbra smtp [mta]—back to postfix from amavis                                                                                                                                                                                                        |
| 10042       | Unofficial |            |      |      | <u>Mathoid</u> server <sup>[345]</sup>                                                                                                                                                                                                               |
| 10050       | Yes        |            |      |      | <u>Zabbix</u> agent                                                                                                                                                                                                                                  |
| 10051       | Yes        |            |      |      | <u>Zabbix</u> trapper                                                                                                                                                                                                                                |
| 10110       | Yes        |            |      |      | NMEA 0183 Navigational Data. Transport of NMEA 0183 sentences over TCP or UDP                                                                                                                                                                        |
| 10172       | Unofficial |            |      |      | Intuit <u>Quickbooks</u> client                                                                                                                                                                                                                      |
| 10200       | Unofficial |            |      |      | FRISK Software International's <i>fpscand</i> virus scanning daemon for Unix platforms <sup>[346]</sup>                                                                                                                                              |
|             | Unofficial |            |      |      | FRISK Software International's <i>f-protd</i> virus scanning daemon for Unix platforms <sup>[347]</sup>                                                                                                                                              |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                      |
|-------------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10201–10204 | Unofficial |            |      |      | FRISK Software International's <i>f-protd</i> virus scanning daemon for Unix platforms <sup>[347]</sup>                                                                          |
| 10212       | Yes        |            |      |      | GE Intelligent Platforms Proficy HMI/SCADA – CIMPLICITY WebView <sup>[348]</sup>                                                                                                 |
| 10308       | Unofficial |            |      |      | <i>Digital Combat Simulator</i> Dedicated Server <sup>[349]</sup>                                                                                                                |
| 10468       |            | Unofficial |      |      | Flyer - discovery protocol                                                                                                                                                       |
| 10480       | Unofficial |            |      |      | <i>SWAT 4</i> Dedicated Server                                                                                                                                                   |
| 10505       |            | Unofficial |      |      | BlueStacks (android simulator) broadcast <sup>[350]</sup>                                                                                                                        |
| 10514       | Unofficial | Unofficial |      |      | TLS-enabled Rsyslog (default by convention)                                                                                                                                      |
| 10578       | Unofficial | No         |      |      | Skyrim Together multiplayer server for The Elder Scrolls V: Skyrim mod.                                                                                                          |
| 10800       | Unofficial |            |      |      | Touhou fight games ( <i>Immaterial and Missing Power</i> , <i>Scarlet Weather Rhapsody</i> , <i>Hisoutensoku</i> , <i>Hopeless Masquerade</i> and <i>Urban Legend in Limbo</i> ) |
| 10823       |            | Unofficial |      |      | <i>Farming Simulator 2011</i>                                                                                                                                                    |
| 10891       | Unofficial |            |      |      | Jungle Disk (this port is opened by the Jungle Disk Monitor service on the localhost)                                                                                            |
| 10933       | Yes        | No         |      |      | Octopus Deploy Tentacle deployment agent <sup>[351]</sup>                                                                                                                        |
| 11001       | Yes        |            |      |      | metasys ( Johnson Controls Metasys java AC control environment )                                                                                                                 |
| 11100       | No         | Unofficial |      |      | <i>Risk of Rain</i> multiplayer server                                                                                                                                           |
| 11111       | Unofficial |            |      |      | RiCcl, Remote Configuration Interface (Redhat Linux)                                                                                                                             |
| 11112       | Yes        |            |      |      | ACR/NEMA Digital Imaging and Communications in Medicine (DICOM)                                                                                                                  |
| 11211       | Unofficial | Unofficial |      |      | <i>memcached</i> <sup>[11]</sup>                                                                                                                                                 |
| 11214       | Unofficial | Unofficial |      |      | memcached incoming SSL proxy                                                                                                                                                     |
| 11215       | Unofficial | Unofficial |      |      | memcached internal outgoing SSL proxy                                                                                                                                            |
| 11235       | Yes        |            |      |      | <i>XCOMPUTE</i> numerical systems messaging (Xplicit Computing) <sup>[352]</sup>                                                                                                 |
| 11311       | Unofficial | Unofficial |      |      | <i>Robot Operating System</i> master                                                                                                                                             |
| 11371       | Yes        |            |      |      | <i>OpenPGP</i> HTTP key server                                                                                                                                                   |
| 11753       | Unofficial |            |      |      | <i>OpenRCT2</i> multiplayer <sup>[353]</sup>                                                                                                                                     |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                              |
|-------------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12000       | Unofficial | Unofficial |      |      | <u>CubeForm</u> , Multiplayer SandBox Game                                                                                                                                                               |
| 12012       |            | Unofficial |      |      | <u>Audition Online Dance Battle</u> , Korea Server—Status/Version Check                                                                                                                                  |
| 12013       | Unofficial | Unofficial |      |      | Audition Online Dance Battle, Korea Server                                                                                                                                                               |
| 12035       |            | Unofficial |      |      | <i>Second Life</i> , used for server UDP in-bound <sup>[354]</sup>                                                                                                                                       |
| 12043       | Unofficial |            |      |      | <i>Second Life</i> , used for LSL HTTPS in-bound <sup>[355]</sup>                                                                                                                                        |
| 12046       | Unofficial |            |      |      | <i>Second Life</i> , used for LSL HTTP in-bound <sup>[355]</sup>                                                                                                                                         |
| 12201       | Unofficial | Unofficial |      |      | Graylog Extended Log Format (GELF) <sup>[356]</sup>                                                                                                                                                      |
| 12222       |            | Yes        |      |      | Light Weight Access Point Protocol (LWAPP) LWAPP data (RFC 5412)                                                                                                                                         |
| 12223       |            | Yes        |      |      | Light Weight Access Point Protocol (LWAPP) LWAPP control (RFC 5412)                                                                                                                                      |
| 12307       |            | Unofficial |      |      | Makerbot UDP Broadcast (client to printer) (JSON-RPC) <sup>[357]</sup>                                                                                                                                   |
| 12308       |            | Unofficial |      |      | Makerbot UDP Broadcast (printer to client) (JSON-RPC) <sup>[357]</sup>                                                                                                                                   |
| 12345       | Unofficial | Unofficial |      |      | <i>Cube World</i> <sup>[358]</sup>                                                                                                                                                                       |
|             | Unofficial |            |      |      | <i>Little Fighter 2</i>                                                                                                                                                                                  |
|             | Unofficial |            |      |      | NetBus remote administration tool (often Trojan horse).                                                                                                                                                  |
| 12443       | Unofficial |            |      |      | IBM HMC web browser management access over <u>HTTPS</u> instead of default port 443 <sup>[359]</sup>                                                                                                     |
| 12489       | Unofficial |            |      |      | NSClient/NSClient++/NC_Net (Nagios)                                                                                                                                                                      |
| 12975       | Unofficial |            |      |      | <u>LogMeIn Hamachi</u> (VPN tunnel software; also port 32976)—used to connect to Mediation Server (bibi.hamachi.cc); will attempt to use <u>SSL</u> (TCP port 443) if both 12975 & 32976 fail to connect |
| 13000–13050 |            | Unofficial |      |      | <i>Second Life</i> , used for server UDP in-bound <sup>[360]</sup>                                                                                                                                       |
| 13008       | Unofficial | Unofficial |      |      | <i>Crossfire</i> , a multiplayer online First Person Shooter                                                                                                                                             |
| 13075       | Yes        |            |      |      | Default <sup>[361]</sup> for BMC Software <u>Control-M/Enterprise Manager Corba</u> communication, though often changed during installation                                                              |

| Port  | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                              |
|-------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------|
| 13400 | Yes        |            |      |      | ISO 13400 Road vehicles — Diagnostic communication over Internet Protocol(DoIP)                                                          |
| 13698 | Unofficial |            |      |      | File-Transferer ( <a href="https://github.com/UltraQbik/File-Transferer">https://github.com/UltraQbik/File-Transferer</a> )              |
| 13720 | Yes        |            |      |      | Symantec <u>NetBackup</u> —bprd (formerly VERITAS)                                                                                       |
| 13721 | Yes        |            |      |      | Symantec NetBackup—bpdbm (formerly VERITAS)                                                                                              |
| 13724 | Yes        |            |      |      | Symantec Network Utility—vnetd (formerly VERITAS)                                                                                        |
| 13782 | Yes        |            |      |      | Symantec NetBackup—bpcd (formerly VERITAS)                                                                                               |
| 13783 | Yes        |            |      |      | Symantec VOPIED protocol (formerly VERITAS)                                                                                              |
| 13785 | Yes        |            |      |      | Symantec NetBackup Database—nbdb (formerly VERITAS)                                                                                      |
| 13786 | Yes        |            |      |      | Symantec nomdb (formerly VERITAS)                                                                                                        |
| 14550 |            | Unofficial |      |      | <u>MAVLink</u> Ground Station Port                                                                                                       |
| 14567 |            | Unofficial |      |      | <u>Battlefield 1942</u> and mods                                                                                                         |
| 14652 | Unofficial |            |      |      | Repgen DoxBox reporting tool                                                                                                             |
| 14800 | Unofficial |            |      |      | <u>Age of Wonders III</u> p2p port <sup>[362]</sup>                                                                                      |
| 15000 | Unofficial |            |      |      | <u>psyBNC</u>                                                                                                                            |
|       | Unofficial |            |      |      | <u>Wesnoth</u>                                                                                                                           |
|       | Unofficial |            |      |      | Kaspersky Network Agent <sup>[363]</sup>                                                                                                 |
|       |            | Unofficial |      |      | Teltonika networks remote management system (RMS)                                                                                        |
| 15009 | Unofficial | Unofficial |      |      | Teltonika networks remote management system (RMS)                                                                                        |
| 15010 | Unofficial | Unofficial |      |      | Teltonika networks remote management system (RMS)                                                                                        |
| 15441 | Unofficial |            |      |      | <u>ZeroNet</u> fileserver                                                                                                                |
| 15567 |            | Unofficial |      |      | <i><u>Battlefield Vietnam</u></i> and mods                                                                                               |
| 15345 | Yes        |            |      |      | <u>XPilot</u> Contact                                                                                                                    |
| 15672 | Unofficial | No         |      |      | <u>RabbitMQ</u> management plugin <sup>[364]</sup>                                                                                       |
| 16000 | Unofficial |            |      |      | Oracle WebCenter Content: Imaging (formerly known as Oracle Universal Content Management). Port though often changed during installation |
|       | Unofficial |            |      |      | <u>shroudBNC</u>                                                                                                                         |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                         |
|-------------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 16080       | Unofficial |            |      |      | macOS Server Web (HTTP) service with performance cache <sup>[365]</sup>                                                                             |
| 16200       | Unofficial |            |      |      | Oracle WebCenter Content: Content Server (formerly known as Oracle Universal Content Management). Port though often changed during installation     |
| 16225       | Unofficial |            |      |      | Oracle WebCenter Content: Content Server Web UI. Port though often changed during installation                                                      |
| 16250       | Unofficial |            |      |      | Oracle WebCenter Content: Inbound Refinery (formerly known as Oracle Universal Content Management). Port though often changed during installation   |
| 16261       | Unofficial | Unofficial |      |      | <i>Project Zomboid</i> multiplayer. Additional sequential ports used for each player connecting to server.                                          |
| 16300       | Unofficial |            |      |      | Oracle WebCenter Content: Records Management (formerly known as Oracle Universal Records Management). Port though often changed during installation |
| 16384       |            | Unofficial |      |      | CISCO Default RTP MIN                                                                                                                               |
| 16384-16403 |            | Unofficial |      |      | Real-time Transport Protocol (RTP), RTP Control Protocol (RTCP), used by Apple's iChat for audio and video <sup>[11]</sup>                          |
| 16384-16387 |            | Unofficial |      |      | Real-time Transport Protocol (RTP), RTP Control Protocol (RTCP), used by Apple's FaceTime and Game Center <sup>[11]</sup>                           |
| 16393-16402 |            | Unofficial |      |      | Real-time Transport Protocol (RTP), RTP Control Protocol (RTCP), used by Apple's FaceTime and Game Center <sup>[11]</sup>                           |
| 16403-16472 |            | Unofficial |      |      | Real-time Transport Protocol (RTP), RTP Control Protocol (RTCP), used by Apple's Game Center <sup>[11]</sup>                                        |
| 16400       | Unofficial |            |      |      | Oracle WebCenter Content: Capture (formerly known as Oracle Document Capture). Port though often changed during installation                        |
| 16567       |            | Unofficial |      |      | <i>Battlefield 2</i> and mods                                                                                                                       |
| 16666       | Unofficial | Unofficial |      |      | SITC Port for mobile web traffic                                                                                                                    |
| 16677       | Unofficial | Unofficial |      |      | SITC Port for mobile web traffic                                                                                                                    |
| 17000       |            | Unofficial |      |      | M17 - Digital RF voice and data protocol with Internet (UDP) gateways (reflectors). <sup>[366]</sup>                                                |
| 17011       | Unofficial |            |      |      | <i>Worms</i> multiplayer                                                                                                                            |

| Port  | TCP        | UDP        | SCTP | DCCP | Description                                                                                                             |
|-------|------------|------------|------|------|-------------------------------------------------------------------------------------------------------------------------|
| 17224 | Yes        |            |      |      | Train Realtime Data Protocol (TRDP) Process Data, network protocol used in train communication. <sup>[2][367]</sup>     |
| 17225 | Yes        |            |      |      | Train Realtime Data Protocol (TRDP) Message Data, network protocol used in train communication. <sup>[2][368]</sup>     |
| 17333 | Unofficial |            |      |      | CS Server (CSMS), default binary protocol port                                                                          |
| 17472 | Yes        |            |      |      | Tanium Communication Port                                                                                               |
| 17474 |            | Unofficial |      |      | DMXControl 3 Network Discovery                                                                                          |
| 17475 | Unofficial | Unofficial |      |      | DMXControl 3 Network Broker                                                                                             |
| 17500 | Yes        |            |      |      | <u>Dropbox</u> LanSync Protocol (db-lsp); used to synchronize file catalogs between Dropbox clients on a local network. |
| 17777 | Unofficial | Unofficial |      |      | SITC Port for mobile web traffic                                                                                        |
| 18080 | Unofficial | No         |      |      | <u>Monero</u> P2P network communications                                                                                |
| 18081 | Unofficial | No         |      |      | Monero incoming RPC calls                                                                                               |
| 18091 | Unofficial | Unofficial |      |      | <u>memcached</u> Internal REST HTTPS for SSL                                                                            |
| 18092 | Unofficial | Unofficial |      |      | memcached Internal CAPI HTTPS for SSL                                                                                   |
| 18104 | Yes        |            |      |      | RAD PDF Service                                                                                                         |
| 18200 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, AsiaSoft Thailand Server status/version check                                             |
| 18201 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, AsiaSoft Thailand Server                                                                  |
| 18206 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, AsiaSoft Thailand Server FAM database                                                     |
| 18300 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, AsiaSoft SEA Server status/version check                                                  |
| 18301 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, AsiaSoft SEA Server                                                                       |
| 18306 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, AsiaSoft SEA Server FAM database                                                          |
| 18333 | Unofficial |            |      |      | <u>Bitcoin</u> testnet                                                                                                  |
| 18400 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, KAIZEN Brazil Server status/version check                                                 |
| 18401 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, KAIZEN Brazil Server                                                                      |
| 18505 | Unofficial | Unofficial |      |      | Audition Online Dance Battle R4p3 Server, Nexon Server status/version check                                             |

| Port  | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                    |
|-------|------------|------------|------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18506 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, Nexon Server                                                                                                                                     |
| 18605 | Unofficial | Unofficial |      |      | <u>X-BEAT</u> status/version check                                                                                                                                             |
| 18606 | Unofficial | Unofficial |      |      | <u>X-BEAT</u>                                                                                                                                                                  |
| 18676 | Unofficial | Unofficial |      |      | YouView                                                                                                                                                                        |
| 19000 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, G10/alaplaya Server status/version check                                                                                                         |
|       |            | Unofficial |      |      | <u>JACK</u> sound server                                                                                                                                                       |
| 19001 | Unofficial | Unofficial |      |      | Audition Online Dance Battle, G10/alaplaya Server                                                                                                                              |
| 19132 |            | Unofficial |      |      | <i>Minecraft: Bedrock Edition</i> multiplayer server <sup>[369]</sup>                                                                                                          |
| 19133 |            | Unofficial |      |      | <i>Minecraft: Bedrock Edition</i> IPv6 multiplayer server <sup>[369]</sup>                                                                                                     |
| 19150 | Unofficial | Unofficial |      |      | <u>Gkrellm</u> Server                                                                                                                                                          |
| 19226 | Unofficial |            |      |      | <u>Panda Software AdminSecure</u> Communication Agent                                                                                                                          |
| 19294 | Unofficial |            |      |      | <u>Google Talk Voice and Video</u> connections <sup>[370]</sup>                                                                                                                |
| 19295 |            | Unofficial |      |      | <u>Google Talk Voice and Video</u> connections <sup>[370]</sup>                                                                                                                |
| 19302 |            | Unofficial |      |      | <u>Google Talk Voice and Video</u> connections <sup>[370]</sup>                                                                                                                |
| 19531 | Unofficial | No         |      |      | <u>systemd-journal-gatewayd</u> <sup>[371]</sup>                                                                                                                               |
| 19532 | Unofficial | No         |      |      | <u>systemd-journal-remote</u> <sup>[372]</sup>                                                                                                                                 |
| 19788 | No         | Yes        |      |      | Mesh Link Establishment protocol for IEEE 802.15.4 radio mesh networks <sup>[373]</sup>                                                                                        |
| 19812 | Yes        | No         |      |      | 4D database SQL Communication <sup>[374]</sup>                                                                                                                                 |
| 19813 | Yes        |            |      |      | 4D database Client Server Communication <sup>[374]</sup>                                                                                                                       |
| 19814 | Yes        |            |      |      | 4D database DB4D Communication <sup>[374]</sup>                                                                                                                                |
| 19999 | Yes        |            |      |      | Distributed Network Protocol—Secure ( <u>DNP</u> —Secure), a secure version of the protocol used in <u>SCADA</u> systems between communicating <u>RTU</u> 's and <u>IED</u> 's |
| 20000 | Yes        |            |      |      | Distributed Network Protocol (DNP), a protocol used in SCADA systems between communicating <u>RTU</u> 's and <u>IED</u> 's                                                     |
|       | Yes        |            |      |      | <u>OpenWebNet</u> , communications protocol used in <u>Bticino</u> products                                                                                                    |

| Port  | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                                                          |
|-------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Unofficial |            |      |      | Usermin, Web-based Unix/Linux user administration tool (default port)                                                                                                                                                                                |
|       | Unofficial |            |      |      | Used on <u>VoIP</u> networks for receiving and transmitting voice telephony traffic which includes <u>Google Voice</u> via the <u>OBiTalk ATA</u> devices as well as on the <u>MagicJack</u> and <u>Vonage</u> ATA network devices. <sup>[344]</sup> |
| 20560 | Unofficial | Unofficial |      |      | <i>Killing Floor</i>                                                                                                                                                                                                                                 |
| 20582 |            | Unofficial |      |      | HW Development IoT comms                                                                                                                                                                                                                             |
| 20583 |            | Unofficial |      |      | HW Development IoT comms                                                                                                                                                                                                                             |
| 20595 |            | Unofficial |      |      | <i>0 A.D. Empires Ascendant</i>                                                                                                                                                                                                                      |
| 20808 |            | Unofficial |      |      | Ableton Link                                                                                                                                                                                                                                         |
| 21025 | Unofficial |            |      |      | Starbound Server (default), <u>Starbound</u> ( <a href="http://playstarbound.com/">http://playstarbound.com/</a> )                                                                                                                                   |
| 21064 | Unofficial |            |      |      | Default Ingres DBMS server                                                                                                                                                                                                                           |
| 22000 | Unofficial |            |      |      | <u>Syncthing</u> (default)                                                                                                                                                                                                                           |
| 22136 | Unofficial |            |      |      | <u>FLIR Systems</u> ( <a href="http://www.flir.com/">http://www.flir.com/</a> )<br>Camera Resource Protocol                                                                                                                                          |
| 22222 | Unofficial |            |      |      | Davis Instruments, <u>WeatherLink IP</u> ( <a href="http://davisnet.com/weather/products/weather_product.asp?pnum=06555">http://davisnet.com/weather/products/weather_product.asp?pnum=06555</a> )                                                   |
| 23073 | Unofficial |            |      |      | <u>Soldat</u> Dedicated Server                                                                                                                                                                                                                       |
| 23399 | Unofficial |            |      |      | <u>Skype</u> default protocol                                                                                                                                                                                                                        |
| 23513 | Unofficial |            |      |      | <i>Duke Nukem 3D</i> source ports                                                                                                                                                                                                                    |
| 24441 | Unofficial | Unofficial |      |      | Pyzor spam detection network                                                                                                                                                                                                                         |
| 24444 | Unofficial |            |      |      | <u>NetBeans</u> integrated development environment                                                                                                                                                                                                   |
| 24465 | Yes        |            |      |      | <u>Tonido Directory Server for Tonido</u> ( <a href="http://www.tonido.com/">http://www.tonido.com/</a> ) which is a Personal Web App and P2P platform                                                                                               |
| 24554 | Yes        |            |      |      | <u>BINKP</u> , <u>Fidonet</u> mail transfers over <u>TCP/IP</u>                                                                                                                                                                                      |
| 24800 | Unofficial |            |      |      | <u>Synergy</u> : keyboard/mouse sharing software                                                                                                                                                                                                     |
| 24842 | Unofficial |            |      |      | <i>StepMania: Online: Dance Dance Revolution</i> Simulator                                                                                                                                                                                           |
| 25565 | Unofficial |            |      |      | <i>Minecraft</i> (Java Edition) multiplayer server <sup>[375]</sup> <sup>[376]</sup>                                                                                                                                                                 |
|       |            | Unofficial |      |      | <i>Minecraft</i> (Java Edition) multiplayer server query <sup>[377]</sup>                                                                                                                                                                            |
| 25575 |            | Unofficial |      |      | <i>Minecraft</i> (Java Edition) multiplayer server RCON <sup>[378]</sup>                                                                                                                                                                             |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                                     |
|-------------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25600-25700 | Unofficial | Unofficial |      |      | SamsidParty Operational Ports                                                                                                                                                                                   |
| 25734-25735 | Unofficial | Unofficial |      |      | SOLIDWORKS SolidNetworkLicense Manager <sup>[379]</sup>                                                                                                                                                         |
| 25826       |            | Unofficial |      |      | <u>collectd</u> default port <sup>[380]</sup>                                                                                                                                                                   |
| 26000       | Yes        | Yes        |      |      | id Software's <u>Quake</u> server                                                                                                                                                                               |
|             | Unofficial |            |      |      | <u>EVE Online</u>                                                                                                                                                                                               |
|             |            | Unofficial |      |      | <u>Xonotic</u> , an <u>open-source</u> arena shooter                                                                                                                                                            |
| 26822       |            | Unofficial |      |      | <u>MSI MysticLight</u>                                                                                                                                                                                          |
| 26900-26901 | Unofficial |            |      |      | <u>EVE Online</u>                                                                                                                                                                                               |
| 26909-26911 | Unofficial |            |      |      | <u>Action Tanks Online</u>                                                                                                                                                                                      |
| 27000       | Unofficial |            |      |      | PowerBuilder <u>SySAM</u> license server                                                                                                                                                                        |
| 27000-27006 |            | Unofficial |      |      | id Software's <u>QuakeWorld</u> master server                                                                                                                                                                   |
| 27000-27009 | Yes        | Yes        |      |      | FlexNet Publisher's License server (from the range of default ports)                                                                                                                                            |
| 27000-27015 | No         | Unofficial |      |      | <u>Steam</u> (game client traffic) <sup>[381]</sup>                                                                                                                                                             |
| 27015       | No         | Unofficial |      |      | GoldSrc and Source engine dedicated server port <sup>[381]</sup>                                                                                                                                                |
| 27015-27018 |            | Unofficial |      |      | <u>Unturned</u> , a survival game                                                                                                                                                                               |
| 27015-27030 | No         | Unofficial |      |      | Steam (matchmaking and HLTV) <sup>[381]</sup>                                                                                                                                                                   |
|             | Unofficial | Unofficial |      |      | Steam (downloads) <sup>[381]</sup>                                                                                                                                                                              |
| 27016       | Unofficial |            |      |      | <u>Magicka</u> and <u>Space Engineers</u> server port                                                                                                                                                           |
| 27017       | Unofficial | No         |      |      | <u>MongoDB</u> daemon process (mongod) and routing service (mongos) <sup>[382]</sup>                                                                                                                            |
| 27031-27035 | No         | Unofficial |      |      | Steam (In-Home Streaming) <sup>[381]</sup>                                                                                                                                                                      |
| 27036       | Unofficial | Unofficial |      |      | Steam (In-Home Streaming) <sup>[381]</sup>                                                                                                                                                                      |
| 27374       | Unofficial |            |      |      | <u>Sub7</u> default.                                                                                                                                                                                            |
| 27500-27900 |            | Unofficial |      |      | id Software's <u>QuakeWorld</u>                                                                                                                                                                                 |
| 27888       |            | Unofficial |      |      | <u>Kaillera</u> server                                                                                                                                                                                          |
| 27901-27910 |            | Unofficial |      |      | id Software's <u>Quake II</u> master server                                                                                                                                                                     |
| 27950       |            | Unofficial |      |      | <u>OpenArena</u> outgoing                                                                                                                                                                                       |
| 27960-27969 |            | Unofficial |      |      | Activision's <u>Enemy Territory</u> and id Software's <u>Quake III Arena</u> , <u>Quake III</u> and <u>Quake Live</u> and some ioquake3 derived games, such as <u>Urban Terror</u> ( <u>OpenArena</u> incoming) |
| 28000       | Yes        | Yes        |      |      | Siemens Digital Industries Software license server <sup>[2]</sup>                                                                                                                                               |

| Port         | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                                  |
|--------------|------------|------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28001        | Unofficial |            |      |      | <i>Starsiege: Tribes</i>                                                                                                                                                                     |
| 28015        |            | Unofficial |      |      | <i>Rust</i> (video game) <sup>[383]</sup>                                                                                                                                                    |
| 28016        |            | Unofficial |      |      | <i>Rust</i> (video game) RCON <sup>[384]</sup>                                                                                                                                               |
| 28260        | Unofficial |            |      |      | Palo Alto Networks' Panorama HA-1 backup unencrypted sync port. <sup>[26]</sup>                                                                                                              |
| 28443        | Unofficial |            |      |      | Palo Alto Networks' Panorama-to-managed devices software updates, PAN-OS 8.0 and later. <sup>[198]</sup>                                                                                     |
| 28769        | Unofficial |            |      |      | Palo Alto Networks' Panorama HA unencrypted sync port. <sup>[26]</sup>                                                                                                                       |
| 28770        | Unofficial |            |      |      | Palo Alto Networks' Panorama HA-1 backup sync port. <sup>[26]</sup>                                                                                                                          |
| 28770–28771  |            | Unofficial |      |      | <i>AssaultCube Reloaded</i> , a video game based upon a modification of <i>AssaultCube</i>                                                                                                   |
| 28785–28786  |            | Unofficial |      |      | <i>Cube 2: Sauerbraten</i> <sup>[385]</sup>                                                                                                                                                  |
| 28852        | Unofficial | Unofficial |      |      | <i>Killing Floor</i>                                                                                                                                                                         |
| 28910        | Unofficial | Unofficial |      |      | Nintendo Wi-Fi Connection <sup>[386]</sup>                                                                                                                                                   |
| 28960        | Unofficial | Unofficial |      |      | <i>Call of Duty</i> ; <i>Call of Duty: United Offensive</i> ; <i>Call of Duty 2</i> ; <i>Call of Duty 4: Modern Warfare</i> <i>Call of Duty: World at War</i> (PC platform) <sup>[387]</sup> |
| 29000        | Unofficial |            |      |      | <i>Perfect World</i> , an adventure and fantasy MMORPG                                                                                                                                       |
| 29070        | Unofficial | Unofficial |      |      | <i>Jedi Knight: Jedi Academy</i> by Ravensoft                                                                                                                                                |
| 29900–29901  | Unofficial | Unofficial |      |      | Nintendo Wi-Fi Connection <sup>[386]</sup>                                                                                                                                                   |
| 29920        | Unofficial | Unofficial |      |      | Nintendo Wi-Fi Connection <sup>[386]</sup>                                                                                                                                                   |
| 30000        |            | Unofficial |      |      | XLink Kai P2P                                                                                                                                                                                |
|              |            | Unofficial |      |      | Minetest server default port <sup>[388]</sup>                                                                                                                                                |
|              |            | Unofficial |      |      | Foundry Virtual Tabletop server default port <sup>[389]</sup>                                                                                                                                |
| 30033        | Unofficial | No         |      |      | TeamSpeak 3 File Transfer <sup>[342]</sup>                                                                                                                                                   |
| 30120        | Unofficial |            |      |      | <i>Fivem</i> (Default Port) GTA V multiplayer <sup>[390][376]</sup>                                                                                                                          |
| 30564        | Unofficial |            |      |      | Multiplicity: keyboard/mouse/clipboard sharing software                                                                                                                                      |
| <u>31337</u> | Unofficial |            |      |      | Back Orifice and Back Orifice 2000 remote administration tools <sup>[391][392]</sup>                                                                                                         |
|              | Unofficial | Unofficial |      |      | ncat, a netcat alternative <sup>[393]</sup>                                                                                                                                                  |
| 31416        | Unofficial |            |      |      | BOINC RPC <sup>[394]</sup>                                                                                                                                                                   |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                     |
|-------------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------|
| 31438       | Unofficial |            |      |      | <u>Rocket U2</u> <sup>[395]</sup>                                                                               |
| 31457       | Yes        |            |      |      | <u>TetriNET</u>                                                                                                 |
| 32137       | Unofficial | Unofficial |      |      | Immunet Protect (UDP in version 2.0, <sup>[396]</sup> TCP since version 3.0 <sup>[397]</sup> )                  |
| 32400       | Yes        |            |      |      | <u>Plex Media Server</u> <sup>[398]</sup>                                                                       |
| 32764       | Unofficial |            |      |      | A backdoor found on certain Linksys, Netgear and other wireless DSL modems/combination routers <sup>[399]</sup> |
| 32887       | Unofficial |            |      |      | <i>Ace of Spades</i> , a multiplayer FPS video game                                                             |
| 32976       | Unofficial |            |      |      | LogMeIn Hamachi, a VPN application; also TCP port 12975 and <u>SSL</u> (TCP 443). <sup>[400]</sup>              |
| 33434       | Yes        | Yes        |      |      | <u>traceroute</u>                                                                                               |
| 33848       |            | Unofficial |      |      | Jenkins, a continuous integration (CI) tool <sup>[401][402]</sup>                                               |
| 34000       |            | Unofficial |      |      | <i>Infestation: Survivor Stories</i> (formerly known as <i>The War Z</i> ), a multiplayer zombie video game     |
| 34197       | No         | Unofficial |      |      | <i>Factorio</i> , a multiplayer survival and factory-building game <sup>[403]</sup>                             |
| 35357       | Yes        |            |      |      | OpenStack Identity (Keystone) administration <sup>[404]</sup>                                                   |
| 36330       | Unofficial |            |      |      | <u>Folding@home</u> Control Port                                                                                |
| 37008       |            | Unofficial |      |      | <u>TZSP</u> intrusion detection                                                                                 |
| 40000       | Yes        | Yes        |      |      | SafetyNET p – a real-time <u>Industrial Ethernet</u> protocol                                                   |
| 41121       | Yes        | Yes        |      |      | Tentacle Server <sup>[405]</sup> - <u>Pandora FMS</u>                                                           |
| 41794       | Yes        | Yes        |      |      | Crestron Control Port <sup>[406]</sup> - <u>Crestron Electronics</u>                                            |
| 41795       | Yes        | Yes        |      |      | Crestron Terminal Port <sup>[407]</sup> - <u>Crestron Electronics</u>                                           |
| 41796       | Yes        | No         |      |      | Crestron Secure Control Port <sup>[408]</sup> - <u>Crestron Electronics</u>                                     |
| 41797       | Yes        | No         |      |      | Crestron Secure Terminal Port <sup>[409]</sup> - <u>Crestron Electronics</u>                                    |
| 42081-42090 | Yes        | Yes        |      |      | Zippin - <u>Zippin Stores</u> ( <a href="https://getzippin.com">https://getzippin.com</a> )                     |
| 42590-42595 | Yes        | Yes        |      |      | Glue - <u>MakePro X</u> ( <a href="https://makepro-x.com">https://makepro-x.com</a> )                           |
| 42806       |            |            |      |      | <u>Discord</u> <sup>[410]</sup>                                                                                 |
| 42999       | Yes        |            |      |      | <u>Curiosity</u> ( <a href="https://curiosity.ai/">https://curiosity.ai/</a> ) <sup>[411]</sup>                 |

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                             |
|-------------|------------|------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 43110       | Unofficial |            |      |      | <u>ZeroNet</u> web UI default port <sup>[412]</sup>                                                                                     |
| 43594–43595 | Unofficial |            |      |      | <u>RuneScape</u> <sup>[413]</sup>                                                                                                       |
| 44405       | Unofficial |            |      |      | <u>Mu Online</u> Connect Server                                                                                                         |
| 44818       | Yes        | Yes        |      |      | <u>EtherNet/IP</u> explicit messaging                                                                                                   |
| 47808–47823 | Yes        | Yes        |      |      | BACnet Building Automation and Control Networks (47808 <sub>10</sub> = BAC0 <sub>16</sub> to 47823 <sub>10</sub> = BACF <sub>16</sub> ) |
| 48556       | Yes        | Yes        |      |      | <u>drive.web</u> AC/DC Drive Automation and Control Networks <sup>[414]</sup>                                                           |
| 48656       | Unofficial |            |      |      | <u>Brainy LAB</u> ( <a href="https://brainy-lab.com/">https://brainy-lab.com/</a> ) Control Server                                      |
| 48657       |            | Unofficial |      |      | <u>Brainy LAB</u> ( <a href="https://brainy-lab.com/">https://brainy-lab.com/</a> ) Control Server                                      |
| 49151       | Reserved   | Reserved   |      |      | Reserved <sup>[2]</sup>                                                                                                                 |

## Dynamic, private or ephemeral ports

The range 49152–65535 ( $2^{15} + 2^{14}$  to  $2^{16} - 1$ ) contains dynamic or private ports that cannot be registered with IANA.<sup>[415]</sup> This range is used for private or customized services, for temporary purposes, and for automatic allocation of ephemeral ports.

Dynamic, private or ephemeral ports <sup>[hide]</sup>

| Port        | TCP        | UDP        | SCTP | DCCP | Description                                                                                                                                                                        |
|-------------|------------|------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 49152–65535 | Unofficial | No         |      |      | <u>Certificate Management over CMS</u> <sup>[416]</sup>                                                                                                                            |
| 49160       | Unofficial |            |      |      | Palo Alto Networks' Panorama. <sup>[198]</sup>                                                                                                                                     |
| 51820       | No         | Unofficial |      |      | <u>WireGuard</u> protocol <sup>[417]</sup>                                                                                                                                         |
| 60000–61000 | No         | Unofficial |      |      | Range from which Mosh – a remote-terminal application similar to <u>SSH</u> – typically assigns ports for ongoing sessions between Mosh servers and Mosh clients. <sup>[418]</sup> |
| 64738       | Unofficial | Unofficial |      |      | <u>Mumble</u> <sup>[419]</sup>                                                                                                                                                     |

## Note

1. TCP port 465 was originally assigned to allow the use of SMTP over SSL (SMTPS), but practical concerns meant that it was left unused and according to the registration rules at that time was subsequently revoked and eventually re-assigned for use by Cisco's URD protocol. Subsequently, port 587 was assigned as the SMTP submission port, but was initially in plaintext, with encryption eventually provided years later by the STARTTLS extension. At the same time, the subsequent adoption of the usage of 465 as an SSL-enabled SMTP submission port, even though that the original

registration did not envision that usage and despite the fact that it was registered to another service has endured. Subsequently, RFC 8314, in a special exemption to the normal assignment process as defined by RFC 6335, has acknowledged the *de-facto* situation and has designated SMTP over TLS as an 'alternate usage assignment'.

2. Deployment typically occurs only directly over UDP, but other underlying protocol layers which meet the requirements described in the specification are possible.

## See also

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- [Port \(computer networking\)](#)
- [Internet protocol suite](#)
- [List of IP numbers](#)
- [Lists of network protocols](#)
- [Comparison of file transfer protocols](#)

[420]

## References and notes

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```
sails lift
```

Your app will be running on localhost:1337. Let's test it. ...

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## External links

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